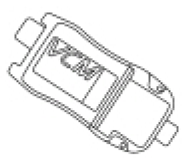
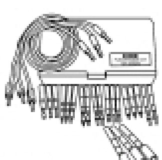
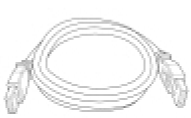
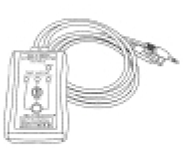
# Information and Entertainment System

Ford Ecat Electronic Parts Catalog

Special Tool(s)

| | |
|---|---|
|  ST3093-A | Fluke 77-IV Digital Multimeter<br>FLU77-4 or equivalent |
|  ST2834-A | Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware or equivalent diagnostic scan tool |
|  ST2574-A | Flex Probe Kit<br>NUD105-R025D or equivalent |
|  ST3104-A | Back-probe Pins<br>POM6411 or equivalent |
|  ST3051-A | Universal Serial Bus (USB) Male-A To Male-A Cable<br>CCMUSB2-AM-AM-10 |
|  ST3086-A | Multi-Media Interface Tester<br>105-00120 |

**DTC Charts**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

**Audio Front Control Module (ACM) DTC Chart**

| DTC | Description | Action |
|---|---|---|
| B1A01:01 | Speaker #1: General Electrical Failure | • For a LF door speaker concern, GO to Pinpoint Test F<br>• For a LF tweeter speaker concern, GO to Pinpoint Test G |
| B1A01:11 | Speaker #1: Circuit Short to Ground | • For a LF door speaker concern, GO to Pinpoint Test F<br>• For a LF tweeter speaker concern, GO to Pinpoint Test G |
| B1A01:12 | Speaker #1: Circuit Short to Battery | • For a LF door speaker concern, GO to Pinpoint Test F<br>• For a LF tweeter speaker concern, GO to Pinpoint Test G |
| B1A01:13 | Speaker #1: Circuit Open | • For a LF door speaker concern, GO to Pinpoint Test F<br>• For a LF tweeter speaker concern, GO to Pinpoint Test G |
| B1A02:01 | Speaker #2: General Electrical Failure | • For a RF door speaker concern, GO to Pinpoint Test H<br>• For a RF tweeter speaker concern, GO to Pinpoint Test I |
| B1A02:11 | Speaker #2: Circuit Short to Ground | • For a RF door speaker concern, GO to Pinpoint Test H<br>• For a RF tweeter speaker concern, GO to Pinpoint Test I |
| B1A02:12 | Speaker #2: Circuit Short to Battery | • For a RF door speaker concern, GO to Pinpoint Test H<br>• For a RF tweeter speaker concern, GO to Pinpoint Test I |
| B1A02:13 | Speaker #2: Circuit Open | • For a RF door speaker concern, GO to Pinpoint Test H<br>• For a RF tweeter speaker concern, GO to Pinpoint Test I |
| B1A03:01 | Speaker #3: General Electrical Failure | GO to Pinpoint Test J. |
| B1A03:11 | Speaker #3: Circuit Short to Ground | GO to Pinpoint Test J. |
| B1A03:12 | Speaker #3: Circuit Short to Battery | GO to Pinpoint Test J. |
| B1A03:13 | Speaker #3: Circuit Open | GO to Pinpoint Test J. |
| B1A04:01 | Speaker #4: General Electrical Failure | GO to Pinpoint Test K. |
| B1A04:11 | Speaker #4: Circuit Short to Ground | GO to Pinpoint Test K. |
| B1A04:12 | Speaker #4: Circuit Short to Battery | GO to Pinpoint Test K. |
| B1A04:13 | Speaker #4: Circuit Open | GO to Pinpoint Test K. |
| B1A56:21 | Antenna: Signal Amplitude < Minimum | If DTC B1A56:21 sets after running the bezel diagnostic tests, disregard this DTC. This test is used for end of line or repair bay testing at the Ford plant. Each plant has its own assigned frequency and signal strength requirements. The running and passing of this test depends on local radio stations. If no local station |

| DTC | Description | Action |
|---|---|---|
| | | is broadcasting on the required frequency, the test will fail. Therefore, tests ran from the bezel diagnostic screen and the DTC generated from the test can be disregarded. |
| B1A89:01 | Satellite Antenna: General Electrical Failure | GO to Pinpoint Test B. |
| B1A89:13 | Satellite Antenna: Circuit Open | GO to Pinpoint Test B. |
| B1D19:49 | Compact Disc Unit: Internal Electronic Failure | • CLEAR the DTC.<br>• WAIT at least 30 seconds.<br>• REPEAT the self-test.<br>• If DTC B1D19:49 is retrieved again, INSTALL a new ACM . REFER to Audio Control Module (ACM). |
| U0140:00 | Lost Communication With Body Control Module: No Sub Type Information | GO to Pinpoint Test AC. |
| U0155:00 | Lost Communication With Instrument Panel Cluster (IPC ) Control Module: No Sub Type Information | GO to Pinpoint Test AE. |
| U0159:00 | Lost Communication With Parking Assist Control Module "A": No Sub Type Information | GO to Pinpoint Test AF. |
| U0197:00 | Lost Communication With Telephone Control Module: No Sub Type Information | GO to Pinpoint Test AK. |
| U0255:00 | Lost Communication With Front Display Interface Module: No Sub Type Information | GO to Pinpoint Test AL. |
| U0256:00 | Lost Communication With Front Controls Interface Module "A": No Sub Type Information | GO to Pinpoint Test AM. |
| U2014:41 | Control Module Hardware: General Checksum Failure | CLEAR the DTCs. REPEAT the self-test. If DTC U2014:41 is retrieved again, INSTALL a new ACM . REFER to Audio Control Module (ACM). |
| U2014:42 | Control Module Hardware: General Memory Failure | CLEAR the DTCs. REPEAT the self-test. If DTC U2014:42 is retrieved again, INSTALL a new ACM . REFER to Audio Control Module (ACM). |
| U2014:96 | Control Module Hardware: Component Internal Failure | CLEAR the DTCs. REPEAT the self-test. If DTC U2014:96 is retrieved again, INSTALL a new ACM . REFER to Audio Control Module (ACM). |
| U201A:51 | Control Module Main Calibration Data: Not Programmed | CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01.<br><br>If there have been no recent service actions, INSTALL a new ACM to correct the failure to retain configuration data. REFER to Audio Control Module (ACM). |

| DTC | Description | Action |
|---|---|---|
| U2100:00 | Initial Configuration Not Complete: No Sub Type Information | CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01.<br><br>If there have been no recent service actions, INSTALL a new ACM to correct the failure to retain configuration data. REFER to Audio Control Module (ACM). |
| U2101:00 | Control Module Configuration Incompatible: No Sub Type Information | CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01.<br><br>If there have been no recent service actions, INSTALL a new ACM to correct the failure to retain configuration data. REFER to Audio Control Module (ACM). |
| U3003:16 | Battery Voltage: Circuit Voltage Below Threshold | GO to Pinpoint Test AN. |
| U3003:17 | Battery Voltage: Circuit Voltage Above Threshold | GO to Pinpoint Test AO. |

**Accessory Protocol Interface Module (APIM) DTC Chart (If Equipped)**

| DTC | Description | Action |
|---|---|---|
| B1252:04 | USB Port: System Internal Failure | GO to Pinpoint Test R. |
| B1252:11 | USB Port: Circuit Short to Ground | GO to Pinpoint Test R. |
| B12B8:04 | USB Port #2: System Internal Failure | **NOTE:** *This DTC refers to USB port #2 on the APIM itself, which is not used.*<br><br>• INSPECT the APIM USB port #2 for damage and REMOVE any debris.<br>• CLEAR the DTCs.<br>• WAIT at least 10 seconds.<br>• REPEAT the self-test.<br>• If DTC B12B8:04 is retrieved again, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00. |
| B12B8:11 | USB Port #2: Circuit Short To Ground | **NOTE:** *This DTC refers to USB port #2 on the APIM itself, which is not used.*<br><br>• INSPECT the APIM USB port #2 for damage and REMOVE any debris.<br>• CLEAR the DTCs.<br>• WAIT at least 10 seconds.<br>• REPEAT the self-test.<br>• If DTC B12B8:11 is retrieved again, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00. |
| B1D79:01 | Microphone Input: General Electrical Failure | GO to Pinpoint Test P. |

| DTC | Description | Action |
|---|---|---|
| B1D79:11 | Microphone Input: Circuit Short To Ground | GO to Pinpoint Test P. |
| B1D79:12 | Microphone Input: Circuit Short To Battery | GO to Pinpoint Test P. |
| B1D79:13 | Microphone Input: Circuit Open | GO to Pinpoint Test P. |
| U0140:00 | Lost Communication With Body Control Module: No Sub Type Information | GO to Pinpoint Test AC. |
| U0155:00 | Lost Communication With Instrument Panel Cluster (IPC ) Control Module: No Sub Type Information | GO to Pinpoint Test AE. |
| U0423:00 | Invalid Data Received from Instrument Panel Cluster Control Module: No Sub Type Information | RETRIEVE and FOLLOW DTCs present in the IPC . REFER to Section 413-01. |
| U2100:00 | Initial Configuration Not Complete: No Sub Type Information | This DTC sets due to incomplete or incorrect APIM programming/provisioning. CHECK vehicle service history for recent service actions related to the APIM . If there have been recent service actions with the APIM , REPEAT/PERFORM the APIM programming as directed by the scan tool. REFER to APIM Programming in the Description and Operation, Section 415-00B.<br><br>If there have been no recent service actions, the APIM has failed to retain configuration data. PERFORM the APIM Hardware Test. INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00. |
| U2101:00 | Control Module Configuration Incompatible: No Sub Type Information | This DTC sets due to incomplete or incorrect APIM programming/provisioning. CHECK vehicle service history for recent service actions related to the APIM . If there have been recent service actions with the APIM , REPEAT/PERFORM the APIM programming as directed by the scan tool. REFER to APIM Programming in the Description and Operation, Section 415-00B.<br><br>If there have been no recent service actions, the APIM has failed to retain configuration data. PERFORM the APIM Hardware Test. INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00. |
| U3000:04 | Control Module: System Internal Failure | GO to Pinpoint Test M. |
| U3000:41 | Control Module: General Checksum Failure | • CLEAR the DTC.<br>• WAIT at least 10 seconds.<br>• REPEAT the self-test.<br>• If DTC U3000:41 is retrieved again, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00. |
| U3000:42 | Control Module: General Memory Failure | • CLEAR the DTC.<br>• WAIT at least 10 seconds.<br>• REPEAT the self-test.<br>• If DTC U3000:42 is retrieved again, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00. |
| U3000:88 | Control Module: Bus Off | **NOTE:** *Network communication/message DTCs can result from intermittent concerns such as damaged wiring or low battery voltage occurrences. Additionally, vehicle repair procedures such as module reprogramming often set* |

| DTC | Description | Action |
|---|---|---|
| | | *these DTCs. Replacing a module to resolve a network DTC is unlikely to resolve the concern. The module was unable to communicate on the network at a point in time. The fault is not currently present.*<br><br>To avoid repeat network concerns:<br><br>• INSPECT for damaged or corroded wiring at the APIM .<br>• TEST the vehicle battery. REFER to Charging Systems in Section 414-00.<br>• CLEAR the DTC.<br>• REPEAT the network test with the diagnostic scan tool. |
| U3003:16 | Battery Voltage: Circuit Voltage Below Threshold | GO to Pinpoint Test AN. |
| U3003:17 | Battery Voltage: Circuit Voltage Above Threshold | GO to Pinpoint Test AO. |

**BCM DTC Chart**

| DTC | Description | Action |
|---|---|---|
| B1331:02 | Compass/Mirror Module: General Signal Failure | GO to Pinpoint Test X. |
| B1331:08 | Compass/Mirror Module: Bus Signal/Message Failure | GO to Pinpoint Test X. |
| B1331:49 | Compass/Mirror Module: Internal Electronic Failure | Sets in continuous memory when the BCM detects an internal fault in the compass module. CLEAR the DTCs and test the system for normal operation. If DTC B1331:49 returns, INSTALL a new compass module. Section 501-09. |
| All Other DTCs | - | REFER to Section 419-10. |

**FCIM DTC Chart**

| DTC | Description | Action |
|---|---|---|
| U0140:00 | Lost Communication With Body Control Module: No Sub Type Information | GO to Pinpoint Test AC. |
| U0184:00 | Lost Communication With Radio: No Sub Type Information | GO to Pinpoint Test AI. |
| U2013:63 | Switch Pack: Circuit / Component Protection Time-Out | • CLEAR the DTC.<br>• WAIT at least 10 seconds.<br>• REPEAT the self-test. |

| DTC | Description | Action |
| --- | --- | --- |
| | | • If DTC U2013:63 is retrieved again, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM). |
| U2014:41 | Control Module Hardware: General Checksum Failure | • CLEAR the DTC.<br>• WAIT at least 10 seconds.<br>• REPEAT the self-test.<br>• If DTC U2014:41 is retrieved again, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM). |
| U2014:42 | Control Module Hardware: General Memory Failure | • CLEAR the DTC.<br>• WAIT at least 10 seconds.<br>• REPEAT the self-test.<br>• If DTC U2014:42 is retrieved again, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM). |
| U201A:51 | Control Module Main Calibration Data: Not Programmed | CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01.<br><br>If there have been no recent service actions, INSTALL a new FCIM to correct the failure to retain configuration data. REFER to Front Controls Interface Module (FCIM). |
| U3003:16 | Battery Voltage: Circuit Voltage Below Threshold | GO to Pinpoint Test AN. |
| U3003:17 | Battery Voltage: Circuit Voltage Above Threshold | GO to Pinpoint Test AO. |

**FDIM DTC Chart**

| DTC | Description | Action |
| --- | --- | --- |
| B1318 | Battery Voltage Low | GO to Pinpoint Test Z. |
| B1342 | ECU is Faulted | **NOTE:** *Repair all other DTCs before diagnosing DTC B1342.*<br><br>• CLEAR the DTC.<br>• WAIT at least 10 seconds.<br>• REPEAT the self-test.<br>• If DTC B1342 is retrieved again, INSTALL a new Front Display Interface Module (FDIM) . REFER to Front Display Interface Module (FDIM). |
| B2477 | Module Configuration Failure | CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01. |

| DTC | Description | Action |
|---|---|---|
| | | If there have been no recent service actions, INSTALL a new FDIM to correct the failure to retain configuration data. REFER to Front Display Interface Module (FDIM). |
| U0140 | Lost Communication with Body Control Module (GEM ) | GO to Pinpoint Test AB. |
| U0155 | Lost Communication with Instrument Panel Cluster (IPC ) Control Panel | GO to Pinpoint Test AD. |
| U0164 | Lost Communication With HVAC Control Module (EATC ) | GO to Pinpoint Test AG. |
| U0184 | Lost Communication With Radio (ACM ) | GO to Pinpoint Test AH. |
| U0197 | Lost Communication with Telephone Control Module | GO to Pinpoint Test AJ. |
| U2050 | No Application Present | CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01.<br><br>If there have been no recent service actions, INSTALL a new FDIM to correct the failure to retain configuration data. REFER to Front Display Interface Module (FDIM). |
| U2051 | One or More Calibration Files Missing / Corrupt | CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01.<br><br>If there have been no recent service actions, INSTALL a new FDIM to correct the failure to retain configuration data. REFER to Front Display Interface Module (FDIM). |

**Global Positioning System Module (GPSM) DTC Chart**

| DTC | Description | Action |
|---|---|---|
| U0100:00 | Lost Communication with ECM/PCM "A": No Sub Type Information | GO to Pinpoint Test AA. |
| U0140:00 | Lost Communication With Body Control Module: No Sub Type Information: No Sub Type Information | GO to Pinpoint Test AC. |
| U0155:00 | Lost Communication With Instrument Panel Cluster (IPC ) Control Module: No Sub Type Information | GO to Pinpoint Test AE. |

| DTC | Description | Action |
|---|---|---|
| U0401:00 | Invalid Data Received from ECM/PCM A: No Sub Type Information | RETRIEVE and FOLLOW DTCs present in the PCM . REFER to Section 303-14. |
| U0422:00 | Invalid Data Received From Body Control Module: No Sub Type Information | RETRIEVE and FOLLOW DTCs present in the BCM . REFER to Section 419-10. |
| U0423:00 | Invalid Data Received from Instrument Panel Cluster Control Module: No Sub Type Information | RETRIEVE and FOLLOW DTCs present in the IPC . REFER to Section 413-01. |
| U2100:00 | Initial Configuration Not Complete: No Sub Type Information | CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01.<br><br>If there have been no recent service actions, INSTALL a new Global Positioning System Module (GPSM) to correct the failure to retain configuration data. REFER to Global Positioning System Module (GPSM). |
| U2101:00 | Control Module Configuration Incompatible: No Sub Type Information | CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01.<br><br>If there have been no recent service actions, INSTALL a new Global Positioning System Module (GPSM) to correct the failure to retain configuration data. REFER to Global Positioning System Module (GPSM). |
| U3000:09 | Control Module: Component Failure | • CLEAR the DTC.<br>• WAIT at least 10 seconds.<br>• REPEAT the self-test.<br>• If DTC U3000:09 is retrieved again, PERFORM a battery reset by disconnecting the battery for 1 minute, then reconnecting it.<br>• CLEAR the DTC.<br>• WAIT at least 10 seconds.<br>• REPEAT the self-test.<br>• If DTC U3000:09 is retrieved again, INSTALL a new Global Positioning System Module (GPSM) . REFER to Global Positioning System Module (GPSM). |
| U3000:41 | Control Module: General Checksum Failure | • CLEAR the DTC.<br>• WAIT at least 10 seconds.<br>• REPEAT the self-test.<br>• If DTC U3000:41 is retrieved again, INSTALL a new Global Positioning System Module (GPSM) . REFER to Global Positioning System Module (GPSM). |
| U3000:42 | Control Module: General Memory Failure | • CLEAR the DTC.<br>• WAIT at least 10 seconds.<br>• REPEAT the self-test.<br>• If DTC U3000:42 is retrieved again, INSTALL a new Global Positioning System Module (GPSM) . REFER to Global Positioning System Module (GPSM). |

| DTC | Description | Action |
|---|---|---|
| U3003:16 | Battery Voltage: Circuit Voltage Below Threshold | GO to Pinpoint Test AN. |
| U3003:17 | Battery Voltage: Circuit Voltage Above Threshold | GO to Pinpoint Test AO. |

**IPC DTC Chart**

| DTC | Description | Action |
|---|---|---|
| U0462:82 | Invalid Data Received from Compass Module: Alive/Sequence Counter Incorrect/Not Updated | GO to Pinpoint Test X. |
| All other DTCs | — | REFER to Section 413-01. |

**SCCM DTC Chart**

| DTC | Description | Action |
|---|---|---|
| B1380:09 | Steering Wheel Right Switch Pack: Component Failure | GO to Pinpoint Test T. |
| B1380:11 | Steering Wheel Right Switch Pack: Circuit Short To Ground | GO to Pinpoint Test T. |
| B1380:17 | Steering Wheel Right Switch Pack: Circuit Voltage Above Threshold | GO to Pinpoint Test T. |
| All other DTCs | — | REFER to Section 211-05. |

**Symptom Charts**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Symptom Chart — General Audio System

| Condition | Possible Causes | Action |
|---|---|---|
| • A module does not respond to the diagnostic scan tool | • Fuse<br>• Wiring, terminals, or connectors<br>• Module | • REFER to Section 418-00. |
| • The RH steering wheel controls are inoperative or do not | • Refer to the Pinpoint Test | • GO to Pinpoint Test T. |

| | | |
|---|---|---|
| operate correctly (if equipped) | | |
| • The CD player is inoperative or does not operate correctly | • CD<br>• ACM | • INSPECT the CD for scratches, fingerprints, a loose paper label, incorrect format, or damage. INSERT a known good CD and TEST the system. If the system operates correctly, the concern was caused by a damaged CD. If the system does not operate correctly, INSTALL a new ACM . REFER to Audio Control Module (ACM). |
| • The speed compensated volume does not operate correctly | • Refer to the Pinpoint Test | • GO to Pinpoint Test C. |
| • The compass is inoperative | • Refer to the Pinpoint Test | • GO to Pinpoint Test X. |
| • The compass is inaccurate | • Refer to the Pinpoint Test | • GO to Pinpoint Test Y. |
| • The FCIM illumination is inoperative or does not dim | • Illumination system concern<br>• FCIM | • REFER to Section 413-00. |
| • The audio system does not operate correctly from the FCIM | • Refer to the Pinpoint Test | • GO to Pinpoint Test U. |
| • The audio input jack is inoperative or does not operate correctly | • Refer to the Pinpoint Test | • GO to Pinpoint Test Q. |
| • The FDIM is inoperative or does not operate correctly | • Refer to the Pinpoint Test | • GO to Pinpoint Test V. |
| • One or more FDIM displays are inoperative or do not operate correctly | • Refer to the Pinpoint Test | • GO to Pinpoint Test W. |
| • The FDIM illumination is inoperative or does not dim | • Illumination system concern<br>• FDIM | • REFER to Section 413-00. |

Symptom Chart — Satellite Radio (If Equipped)

| **Condition** | **Possible Causes** | **Action** |
|---|---|---|

| | | |
|---|---|---|
| • Poor reception — satellite radio | • Refer to the Pinpoint Test | • GO to Pinpoint Test B. |
| • Poor sound quality or no sound while in satellite radio mode - all other functions operate correctly | • Refer to the Pinpoint Test | • GO to Pinpoint Test B. |
| • The satellite radio is inoperative or does not operate correctly | • Refer to the Pinpoint Test | • GO to Pinpoint Test B. |

Symptom Chart — Sound Quality

| Condition | Possible Causes | Action |
|---|---|---|
| • Poor reception — AM/FM | • Refer to the Pinpoint Test | • GO to Pinpoint Test A. |
| • Continuous seek or scan — AM/FM | • Refer to the Pinpoint Test | • GO to Pinpoint Test A. |
| • Poor sound quality or distorted sound from one or more speakers (not all speakers) | • Refer to the Pinpoint Test | • GO to Pinpoint Test D. |
| • No sound from all speakers | • Refer to the Pinpoint Test | • GO to Pinpoint Test E. |
| • The LF door speaker is inoperative | • Refer to the Pinpoint Test | • GO to Pinpoint Test F. |
| • The LF tweeter speaker is inoperative | • Refer to the Pinpoint Test | • GO to Pinpoint Test G. |
| • The RF door speaker is inoperative | • Refer to the Pinpoint Test | • GO to Pinpoint Test H. |
| • The RF tweeter speaker is inoperative | • Refer to the Pinpoint Test | • GO to Pinpoint Test I. |
| • The RR door speaker is inoperative | • Refer to the Pinpoint Test | • GO to Pinpoint Test J. |
| • The LR door speaker is inoperative | • Refer to the Pinpoint Test | • GO to Pinpoint Test K. |

Symptom Chart — SYNC® System (If Equipped)

| Condition | Possible Causes | Action |
|---|---|---|
| • All SYNC® functions are inoperative or do not operate correctly | • Refer to the Pinpoint Test | • PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test M. |
| • All SYNC® Services features are inoperative or inaccurate | • Refer to the Pinpoint Test | • PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test O. |

| | | |
|---|---|---|
| • The SYNC® system audible prompts are inoperative or do not operate correctly | • Refer to the Pinpoint Test | • PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test N. |
| • During a phone call, no incoming audio is heard in the vehicle | • Refer to the Pinpoint Test | • PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test N. |
| • Voice recognition is inoperative or does not operate correctly | • Refer to the Pinpoint Test | • PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test P. |
| • During a phone call, poor quality or no outgoing audio is heard on the outside device | • Refer to the Pinpoint Test | • PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test P. |
| • Poor quality, distorted, or no sound while in SYNC® mode | • Refer to the Pinpoint Test | • PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test L. |
| • The RH SYNC® steering wheel controls are inoperative or do not operate correctly | • Refer to the Pinpoint Test | • PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test T. |
| • The audio input jack is inoperative or does not operate correctly (with SYNC®) | • Refer to the Pinpoint Test | • PERFORM the Accessory Protocol Interface Module (APIM) Reset. REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test Q. |
| • The USB port is inoperative or does not operate correctly | • Refer to the Pinpoint Test | • PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test R. |
| • Unable to pair Bluetooth device | • Refer to the Pinpoint Test | • PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test S. |
| • An individual Bluetooth device feature is inoperative (text messaging, media playback or controls, phone book download, call history, or call waiting) | • Customer's device compatibility | • PERFORM the Accessory Protocol Interface Module (APIM) Reset. REFER to Accessory Protocol Interface Module (APIM) Reset.<br>• If the concern is still present, INSTRUCT the customer to review the device compatibility list on the SyncMyRide website. It is normal operation for unsupported or incompatible Bluetooth features to be unavailable through the SYNC® system. |

| | | |
|---|---|---|
| | | • If the inoperative feature is compatible, ADVISE the customer to check their device for firmware updates, and if required, reset their device following the manufacturer's instructions. |
| • SYNC® AppLink™ is inoperative or does not operate correctly | • Refer to the Pinpoint Test | • PERFORM the Accessory Protocol Interface Module (APIM) Reset. REFER to Accessory Protocol Interface Module (APIM) Reset.<br><br>• If the concern is still present, INSTRUCT the customer to review the mobile device platform, application, and application version compatibility list on the SyncMyRide website. It is normal operation for incompatible mobile device platforms, applications, and application versions to be unavailable through SYNC® AppLink™ .<br><br>• If the mobile device platform, application, and application version are compatible, and the device is connected through Bluetooth only, GO to Pinpoint Test S.<br><br>• If the mobile device platform, application, and application version are compatible, and the device is connected through Bluetooth and the USB port, GO to Pinpoint Test R |
| • An individual Bluetooth device feature is inoperative | • Customer's device compatibility | • **NOTE:** *If a Bluetooth device is able to pair with the SYNC® system, there are no concerns with the APIM .*<br><br>• INSTRUCT the customer to review the device compatibility list on the the SyncMyRide website. Not all phone features are available through the SYNC® system. This is normal operation. |
| • Unable to pair Bluetooth device or the Bluetooth functionality is inoperative | • Refer to the Pinpoint Test | • GO to Pinpoint Test S. |
| • SYNC® AppLink™ is inoperative or does not operate correctly | • Refer to the Pinpoint Test | • **NOTE:** *NOTE: SYNC® AppLink™ is compatible with select smartphones and select applications. INSTRUCT the customer to review the device compatibility list. REFER to the SyncMyRide website.*<br><br>• If SYNC® AppLink™ is inoperative through Bluetooth, GO to Pinpoint Test S.<br><br>• If SYNC® AppLink™ is inoperative from the USB port, GO to Pinpoint Test R. |

| | | |
|---|---|---|
| • 911 Assist™ or Vehicle Health Report (VHR) is inoperative or does not operate correctly | • Unregistered SYNC® owner account<br>• RCM misconfiguration<br>• IPC misconfiguration | • Verify that the customer has a SYNC® Owner Account on the SyncMyRide website.<br>• This error message appears due to incomplete or incorrect PMI procedures. CHECK the vehicle service history for recent service actions related to the IPC or RCM .<br>• If there have been recent service actions with the IPC or RCM , REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. REFER to Programmable Module Installation (PMI) in Section 418-01.<br>• If there have been no recent service actions, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00. |

## Pinpoint Tests

### Pinpoint Test A: Poor Reception Or Continuous Seek Or Scan — AM/FM

#### Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

#### Normal Operation and Fault Conditions

Refer to AM/FM Radio, ACM , and Antenna in Information and Entertainment System.

#### Possible Sources

- AM/FM antenna
- AM/FM antenna coaxial cable
- Charging system
- Ignition system
- No channel found in the selected category
- Noise suppression equipment
- Wiring, terminals, or connectors
- ACM

**PINPOINT TEST A : POOR RECEPTION OR CONTINUOUS SEEK OR SCAN — AM/FM**

**A1 CHECK THE AUDIO SYSTEM RECEPTION**

- Operate the audio system in radio tuner AM/FM mode.
- Check the reception with the engine running, and with the engine off.

**Does the poor reception only occur with the engine running?**

| | |
|---|---|
| **Yes** | GO to A2. |
| **No** | GO to A4. |

## A2 CHECK THE GENERATOR

- Ignition OFF.
- Disconnect: Wiring harness to the generator voltage regulator.
- Start the engine.
- Operate the audio system in radio tuner AM/FM mode.

**Is the reception OK?**

| | |
|---|---|
| **Yes** | INSTALL a new generator. REFER to Section 414-00. |
| **No** | GO to A3. |

## A3 CHECK THE IGNITION CIRCUITS

- Ignition OFF.
- Connect: Wiring harness to the generator voltage regulator.
- Visually inspect the engine compartment and make sure all ignition coils are correctly and securely connected, and that there are no visible cracks in the coil housings.
- Inspect all the wiring harnesses and connectors for damaged insulation and loose or broken conditions.

**Are the ignition components OK?**

| | |
|---|---|
| **Yes** | USE a jumper cable to ground various parts of the vehicle to the frame (for example: engine, fenders, quarter panels, stone deflectors, air cleaner, body sheet metal). When the noise is eliminated, PROVIDE a permanent ground where necessary. |
| **No** | REPAIR the ignition system as necessary. |

## A4 CHECK THE LOWER AM/FM ANTENNA COAXIAL CABLE FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: AM/FM antenna coaxial cable at the ACM . Refer to Audio Control Module (ACM).
- Disconnect: AM/FM antenna inline coaxial cable connection behind the glove compartment. Refer to Antenna Cable — AM/FM.
- Ignition ON.
- Measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| AM/FM antenna coaxial cable core at the ACM | — | — | Ground |

| | | | |
|---|---|---|---|
| AM/FM antenna coaxial cable shield at the ACM | — | — | Ground |

**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR or INSTALL a new lower AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM. |
| **No** | GO to A5. |

**A5 CHECK THE LOWER AM/FM ANTENNA COAXIAL CABLE CORE FOR A SHORT TO GROUND**

- Ignition OFF.
- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| AM/FM antenna coaxial cable core at the ACM | — | — | Ground |

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to A6. |
| **No** | REPAIR or INSTALL a new lower AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM. |

**A6 CHECK THE LOWER AM/FM ANTENNA COAXIAL CABLE FOR AN OPEN**

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| AM/FM coaxial cable core at the ACM | — | Lower AM/FM antenna coaxial cable core at the inline connection | — |
| AM/FM antenna coaxial cable shield at the ACM | — | Lower AM/FM antenna coaxial cable shield at the inline connection | — |

**Is the resistance less than 1 ohm?**

| | |
|---|---|
| **Yes** | GO to A7. |

| | |
|---|---|
| **No** | REPAIR or INSTALL a new lower AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM. |

**A7 CHECK THE LOWER AM/FM ANTENNA COAXIAL CABLE CORE AND SHIELD FOR A SHORT TOGETHER**

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| AM/FM antenna coaxial cable core at the ACM | — | AM/FM antenna coaxial cable shield at the ACM | — |

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to A8. |
| **No** | REPAIR or INSTALL a new lower AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM. |

**A8 CHECK THE UPPER AM/FM ANTENNA COAXIAL CABLE FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Connect: AM/FM antenna inline coaxial cable connection behind the glove compartment.
- Disconnect: AM/FM antenna coaxial cable at the antenna unit. Refer to Antenna — AM/FM.
- Ignition ON.
- Measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| AM/FM antenna coaxial cable core at the ACM | — | — | Ground |
| AM/FM antenna coaxial cable shield at the ACM | — | — | Ground |

**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR or INSTALL a new upper AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM. |
| **No** | GO to A9. |

**A9 CHECK THE UPPER AM/FM ANTENNA COAXIAL CABLE CORE FOR A SHORT TO GROUND**

- Ignition OFF.
- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| AM/FM antenna coaxial cable core at the ACM | — | — | Ground |

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to A10. |
| **No** | REPAIR or INSTALL a new upper AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM. |

**A10 CHECK THE UPPER AM/FM ANTENNA COAXIAL CABLE FOR AN OPEN**

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| AM/FM antenna coaxial cable core at the ACM | — | AM/FM antenna coaxial cable core at the antenna unit (harness side) | — |
| AM/FM antenna coaxial cable shield at the ACM | — | AM/FM antenna coaxial cable shield at the antenna unit (harness side) | — |

**Is the resistance less than 1 ohm?**

| | |
|---|---|
| **Yes** | GO to A11. |
| **No** | REPAIR or INSTALL a new upper AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM. |

**A11 CHECK THE UPPER AM/FM ANTENNA COAXIAL CABLE CORE AND SHIELD FOR A SHORT TOGETHER**

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| AM/FM antenna coaxial cable core at the | — | AM/FM antenna coaxial cable shield at the | — |

| | | | |
|---|---|---|---|
| ACM | | ACM | |

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to A12. |
| **No** | REPAIR or INSTALL a new upper AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM. |

**A12 CHECK THE AM/FM ANTENNA UNIT FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Connect: AM/FM antenna coaxial cable at the antenna unit.
- Ignition ON.
- Measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| AM/FM antenna coaxial cable core at the ACM | — | — | Ground |
| AM/FM antenna coaxial cable shield at the ACM | — | — | Ground |

**Is any voltage present?**

| | |
|---|---|
| **Yes** | INSTALL a new AM/FM antenna. REFER to Antenna — AM/FM. |
| **No** | GO to A13. |

**A13 CHECK THE AM/FM ANTENNA UNIT CORE FOR A SHORT TO GROUND**

- Ignition OFF.
- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| AM/FM antenna coaxial cable core at the ACM | — | — | Ground |

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to A14. |
| **No** | INSTALL a new AM/FM antenna. REFER to Antenna — AM/FM. |

**A14 CHECK THE AM/FM ANTENNA UNIT CORE AND SHIELD FOR A SHORT TOGETHER**

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| AM/FM antenna coaxial cable core at the ACM | — | AM/FM antenna coaxial cable shield at the ACM | — |

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to A15. |
| **No** | INSTALL a new AM/FM antenna. REFER to Antenna — AM/FM. |

**A15 CHECK THE AM/FM ANTENNA UNIT SHIELD FOR CONTINUITY TO GROUND**

- Disconnect: AM/FM antenna coaxial cable at the antenna unit.
- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| AM/FM antenna unit shield at the antenna unit (component side) | — | — | Ground |

**Is the resistance less than 1 ohm?**

| | |
|---|---|
| **Yes** | GO to A17. |
| **No** | GO to A16. |

**A16 CHECK THE AM/FM ANTENNA UNIT SHIELD FOR CONTINUITY TO ANTENNA UNIT BASE**

- Remove the AM/FM antenna. Refer to Antenna — AM/FM.
- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |

| | | | |
|---|---|---|---|
| AM/FM antenna unit shield at the antenna unit (component side) | — | AM/FM antenna base (unpainted surface) | — |

**Is the resistance less than 1 ohm?**

| | |
|---|---|
| **Yes** | GO to A17. |
| **No** | INSTALL a new AM/FM antenna. REFER to Antenna — AM/FM. |

**A17 CHECK FOR CORRECT ACM OPERATION**

- Ignition OFF.
- Disconnect and inspect all the ACM connectors.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the ACM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

### Pinpoint Test B: The Satellite Radio Is Inoperative, Has Poor Reception, Or Has No Sound

#### Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

#### Normal Operation and Fault Conditions

Refer to MyKey® and Satellite Radio in Information and Entertainment System.

When a satellite radio subscription is activated, the satellite receiver ESN is associated with the VIN .

## DTC Fault Trigger Conditions

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| B1A89:01 | Satellite Antenna: General Electrical Failure | Set by the ACM when a short to ground is detected in the satellite antenna circuit for more than 250 milliseconds. This DTC can be either continuous or on-demand. |
| B1A89:13 | Satellite Antenna: Circuit Open | Set by the ACM when an open is detected in the satellite antenna circuit. This DTC can be either continuous or on-demand. |

**Possible Sources**

- Expired subscription
- Incorrectly matched ESN
- Wiring, terminals, or connectors
- Obstructions in the antenna's line of sight
- Satellite radio antenna
- Satellite radio antenna coaxial cable
- ACM

### PINPOINT TEST B : THE SATELLITE RADIO IS INOPERATIVE, HAS POOR RECEPTION, OR HAS NO SOUND

**B1 VERIFY AN ACTIVE SUBSCRIPTION**

- Operate the infotainment system in satellite radio mode and observe the infotainment display.

**Does the display indicate that the subscription has expired?**

| | |
|---|---|
| **Yes** | The subscription has expired. INFORM the customer to contact SIRIUS to reactivate the subscription. |
| **No** | GO to B2. |

**B2 CHECK THE RECEPTION**

- Drive the vehicle to an open location that is free from obstacles such as tall buildings or large trees.
- Using a diagnostic scan tool, view the ACM PIDs.
- Monitor the satellite signal strength (SAT_SIG_STR) PID.
- If the PID reads "No Signal" or "Poor," move the vehicle to another area and retest until "Good," "Very Good," or "Excellent" PID values are obtained.

**Can a "Good," "Very Good," or "Excellent" PID value be obtained?**

| | |
|---|---|
| **Yes** | GO to B3. |
| **No** | GO to B6. |

**B3 CHECK THE SATELLITE RADIO SYSTEM FOR CORRECT OPERATION**

- Operate multiple channels with the infotainment system in satellite radio mode.

**Does the satellite radio system receive multiple channels with normal sound quality?**

| | |
|---|---|
| **Yes** | The system is operating correctly at this time. The concern was a result of obstructions in the antenna's line of sight. |
| **No** | GO to B4. |

**B4 CHECK FOR PROMOTIONAL CHANNEL 184**

- Select channel 184.

**Does the satellite radio system receive channel 184 with normal sound quality?**

| | |
|---|---|
| **Yes** | GO to B5. |
| **No** | GO to B6. |

**B5 VERIFY THAT THE VIN AND ESN ON FILE MATCH THE VIN AND ESN FROM THE VEHICLE**

- Retrieve the satellite radio ESN . Satellite Radio Electronic Serial Number (ESN) Retrieval.
- Contact the SIRIUS Dealer Support Line (888-465-8528).
- Ensure that the VIN and ESN SIRIUS has on the customer's file match the VIN and ESN retrieved from the vehicle.
- Follow the actions from step 2 to obtain "Good," "Very Good," or "Excellent" reception.
- Operate the infotainment system in satellite radio mode.
- **NOTE:** *It may take up to 30 minutes for the activation signal to be received and satellite radio functions to operate*.

  Request for SIRIUS to send the activation signal to the vehicle 2-3 times.
- Operate multiple channels with the infotainment system in satellite radio mode.

**Does the satellite radio system receive multiple channels with normal sound quality?**

| | |
|---|---|
| **Yes** | The system is operating correctly at this time. The concern was a result of the activation signal needing to be resent or the VIN and ESN SIRIUS had on file not matching the VIN and ESN from the vehicle. |
| **No** | GO to B6. |

**B6 CHECK FOR ACM DTCS**

- Ignition ON.
- Using a diagnostic scan tool, clear the ACM DTCs.
- Using a diagnostic scan tool, perform the ACM self-test.

**Are any DTCs recorded in the ACM ?**

| | |
|---|---|
| **Yes** | For DTC B1A89:01 or DTC B1A89:13, GO to B7.<br>For all other ACM DTCs, REFER to the ACM DTC chart in Information and Entertainment System. |
| **No** | GO to B20. |

**B7 CHECK THE LOWER SATELLITE RADIO ANTENNA COAXIAL CABLE FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: Satellite radio antenna coaxial cable at the ACM . Refer to Audio Control Module (ACM).
- Disconnect: Satellite radio antenna inline coaxial cable connection behind the glove compartment. Refer to Antenna Cable — Satellite Radio.
- Ignition ON.
- Measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| Satellite radio antenna coaxial cable core at the ACM | — | — | Ground |
| Satellite radio antenna coaxial cable shield at the ACM | — | — | Ground |

**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR or INSTALL a new lower satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio. |
| **No** | GO to B8. |

**B8 CHECK THE LOWER SATELLITE RADIO ANTENNA COAXIAL CABLE CORE FOR A SHORT TO GROUND**

- Ignition OFF.
- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| Satellite radio antenna coaxial cable core at the ACM | — | — | Ground |

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to B9. |
| **No** | REPAIR or INSTALL a new lower satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio. |

**B9 CHECK THE LOWER SATELLITE RADIO ANTENNA COAXIAL CABLE FOR AN OPEN**

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| Satellite radio coaxial cable core at the ACM | — | Lower satellite radio antenna coaxial cable core at the inline connection | — |
| Satellite radio antenna coaxial cable shield at the ACM | — | Lower satellite radio antenna coaxial cable shield at the inline connection | — |

**Is the resistance less than 1 ohm?**

| | |
|---|---|
| **Yes** | GO to B10. |
| **No** | REPAIR or INSTALL a new lower satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio. |

**B10 CHECK THE LOWER SATELLITE RADIO ANTENNA COAXIAL CABLE CORE AND SHIELD FOR A SHORT TOGETHER**

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| Satellite radio antenna coaxial cable core at the ACM | — | Satellite radio antenna coaxial cable shield at the ACM | — |

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to B11. |
| **No** | REPAIR or INSTALL a new lower satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio. |

**B11 CHECK THE UPPER SATELLITE RADIO ANTENNA COAXIAL CABLE FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Connect: Satellite radio antenna inline coaxial cable connection behind the glove compartment.
- Disconnect: Satellite radio antenna coaxial cable at the antenna unit. Refer to Antenna — Satellite Radio.
- Ignition ON.
- Measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| Satellite radio antenna coaxial cable core at the ACM | — | — | Ground |
| Satellite radio antenna coaxial cable shield at the ACM | — | — | Ground |

**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR or INSTALL a new upper satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio. |
| **No** | GO to B12. |

**B12 CHECK THE UPPER SATELLITE RADIO ANTENNA COAXIAL CABLE CORE FOR A SHORT TO GROUND**

- Ignition OFF.
- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| Satellite radio antenna coaxial cable core at the ACM | — | — | Ground |

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to B13. |
| **No** | REPAIR or INSTALL a new upper satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio. |

**B13 CHECK THE UPPER SATELLITE RADIO ANTENNA COAXIAL CABLE FOR AN OPEN**

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| Satellite radio antenna coaxial cable core at the ACM | — | Satellite radio antenna coaxial cable core at the antenna unit (harness side) | — |
| Satellite radio antenna coaxial cable shield at the ACM | — | Satellite radio antenna coaxial cable shield at the antenna unit (harness side) | — |

**Is the resistance less than 1 ohm?**

| | |
|---|---|
| **Yes** | GO to B14. |
| **No** | REPAIR or INSTALL a new upper satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio. |

**B14 CHECK THE UPPER SATELLITE RADIO ANTENNA COAXIAL CABLE CORE AND SHIELD FOR A SHORT TOGETHER**

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| Satellite radio antenna coaxial cable core at the ACM | — | Satellite radio antenna coaxial cable shield at the ACM | — |

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to B15. |
| **No** | REPAIR or INSTALL a new upper satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio. |

**B15 CHECK THE SATELLITE RADIO ANTENNA UNIT FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Connect: Satellite radio antenna coaxial cable at the antenna unit.
- Ignition ON.
- Measure the **voltage** between:

| Positive Lead | Negative Lead |
|---|---|

| | | | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| Satellite radio antenna coaxial cable core at the ACM | — | — | Ground |
| Satellite radio antenna coaxial cable shield at the ACM | — | — | Ground |

**Is any voltage present?**

| | |
|---|---|
| **Yes** | INSTALL a new satellite radio antenna. REFER to Antenna — Satellite Radio. |
| **No** | GO to B16. |

**B16 CHECK THE SATELLITE RADIO ANTENNA UNIT CORE FOR A SHORT TO GROUND**

- Ignition OFF.
- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| Satellite radio antenna coaxial cable core at the ACM | — | — | Ground |

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to B17. |
| **No** | INSTALL a new satellite radio antenna. REFER to Antenna — Satellite Radio. |

**B17 CHECK THE SATELLITE RADIO ANTENNA UNIT CORE AND SHIELD FOR A SHORT TOGETHER**

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| Satellite radio antenna coaxial cable core at the ACM | — | Satellite radio antenna coaxial cable shield at the ACM | — |

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to B18. |
| **No** | INSTALL a new satellite radio antenna. REFER to Antenna — Satellite Radio. |

## B18 CHECK THE SATELLITE RADIO ANTENNA UNIT SHIELD FOR CONTINUITY TO GROUND

- Disconnect: Satellite radio antenna coaxial cable at the antenna unit.
- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| Satellite radio antenna unit shield at the antenna unit (component side) | — | — | Ground |

**Is the resistance less than 1 ohm?**

| | |
|---|---|
| **Yes** | GO to B20. |
| **No** | GO to B19. |

## B19 CHECK THE SATELLITE RADIO ANTENNA UNIT SHIELD FOR CONTINUITY TO ANTENNA UNIT BASE

- Remove the satellite radio antenna. Refer to Antenna — Satellite Radio.
- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| Satellite radio antenna unit shield at the antenna unit (component side) | — | Satellite radio antenna base (unpainted surface) | — |

**Is the resistance less than 1 ohm?**

| | |
|---|---|
| **Yes** | GO to B20. |
| **No** | INSTALL a new satellite radio antenna. REFER to Antenna — Satellite Radio. |

## B20 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the ACM connectors.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)

- Reconnect the ACM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test C: The Speed Compensated Volume Does Not Operate Correctly**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

**Normal Operation and Fault Conditions**

Refer to Speed Compensated Volume in Information and Entertainment System.

**Possible Sources**

- Speed compensated volume setting
- VSS signal concern
- Network communication concern
- ACM

**PINPOINT TEST C : THE SPEED COMPENSATED VOLUME DOES NOT OPERATE CORRECTLY**

**C1 CHECK THE SPEEDOMETER OPERATION**

- Drive the vehicle and observe the speedometer.

**Does the speedometer operate correctly?**

| | |
|---|---|
| **Yes** | GO to C2. |
| **No** | REFER to Section 413-01. |

**C2 CHECK THE SPEED COMPENSATED VOLUME SETTING**

- Turn the speed compensated volume off. Refer to the Owner's Literature.
- Operate the audio system in radio tuner AM/FM mode.
- Drive the vehicle at various speeds and observe the speaker volume.
- Set the speed compensated volume to maximum compensation. Refer to the Owner's Literature.
- Operate the audio system in radio tuner AM/FM mode.

- Drive the vehicle at various speeds and observe the speaker volume.

**Does the volume remain constant with the speed compensated volume turned off, and increase and decrease with vehicle speed with the speed compensated volume set to maximum?**

| | |
|---|---|
| **Yes** | The system is operating correctly at this time. INSTRUCT the customer in the correct usage of the speed compensated volume feature. |
| **No** | GO to C3. |

**C3 CHECK FOR DTC U0155:00**

- Using a diagnostic scan tool, retrieve all CMDTCs .

**Is DTC U0155:00 retrieved in any audio system module?**

| | |
|---|---|
| **Yes** | GO to Pinpoint Test AC. |
| **No** | GO to C4. |

**C4 CHECK FOR CORRECT ACM OPERATION**

- Ignition OFF.
- Disconnect and inspect all the ACM connectors.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the ACM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

## Pinpoint Test D: Poor Sound Quality Or Distorted Sound (Buzz/Crackle/Rattle) From One Or More Speakers (Not All Speakers)

### Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

**Normal Operation and Fault Conditions**

The ACM sends audio signals to the speakers in the form of AC voltage, resulting in clear audio output. REFER to Information and Entertainment System. See USB Audio Mode.

**Possible Sources**

- Loose component fasteners
- Loose door handle, lock, and other trim panel components and fasteners
- Loose items in storage areas
- Loose speaker grilles
- Loose wire harnesses or wire harness fasteners
- Speaker(s)
- Water shield bonding and placement

**Visual Inspection and Diagnostic Pre-checks**

Visually inspect around the suspect area for any possible rattle conditions such as:

- Any storage compartments including cup holders (if equipped)
- Door map pockets for contents that can rattle (if equipped)
- Door pull cup attachments (if equipped)
- Door switch bezels (if equipped)
- Package tray (if equipped)
- Speaker grilles
- Trim panels and fasteners

**PINPOINT TEST D : POOR SOUND QUALITY OR DISTORTED SOUND (BUZZ/CRACKLE/RATTLE) FROM ONE OR MORE SPEAKERS (NOT ALL SPEAKERS)**

**D1 ISOLATE THE ZONE**

- Download the diagnostic sound tracks to a suitable USB drive then install the USB drive into the Media Hub. Refer to Speaker Audio Test.
- On the audio system select USB input.
- Using the audio system fade and balance feature, adjust the audio sound to each of the four zones (LF , RF , LR , and RR ) of the vehicle and use the 6 diagnostic sound tracks to isolate the poor sound quality. Allow each track to run for 10 seconds, then use the Seek buttons to change the test track.
- Locate and apply pressure to the trim panel(s) around the poor sound quality area in question.

**Does applying pressure to a trim panel reduce or eliminate the audible noise?**

| | |
|---|---|
| **Yes** | REPAIR or REPLACE the trim panel as needed. |
| **No** | GO to D2. |

**D2 REMOVE AND INSPECT BEHIND/UNDERNEATH THE SUSPECT TRIM PANEL(S)**

- Remove the trim panel to access the suspect speaker.
- Operate the audio system using digital media (CD, MP3, etc.).
- Validate where the poor sound quality area is located.
- Check:
  - Trim panel for loose components around the speaker area such as the speaker grille

- Trim panel joining components for missing or broken pieces
- Lock and handle mechanical parts for correct attachment
- Wire harnesses for correct routing
- Wire harness fasteners for correct attachment
- Water shield for correct placement (speaker air path not blocked)
- Water shield for correct bonding to sheet metal
- Storage areas for loose items
- All child safety belt anchors (if equipped)
- All safety belt retractors
- All speaker bracket fasteners and other fasteners are secured and tightened to specified torque

**Was the source of the noise located?**

| | |
|---|---|
| **Yes** | REPAIR or REPLACE any loose or broken component or fastener as needed. |
| **No** | GO to D3. |

**D3 CHECK THE SUSPECT SPEAKER FOR WATER INTRUSION**

- Inspect for watermarks.
- Check the:
  - Cone
  - Magnet
  - Basket

**Are any watermarks present on the speaker?**

| | |
|---|---|
| **Yes** | VERIFY the water shield is in the correct location, REPAIR or REPLACE any trim, door, or speaker seal as required.<br>DRY the speaker in question and TEST the system for normal operation. |
| **No** | GO to D4. |

**D4 ISOLATE SPEAKER TO VERIFY NOISE**

- Remove the suspect speaker from its location and leave the speaker connected to the harness.
- Hold the speaker away from any trim panel(s) and ensure the speaker is isolated from contact.
- Operate the audio system using digital media (CD, MP3, etc.).

**Is the noise still present in the suspect speaker?**

| | |
|---|---|
| **Yes** | INSTALL a new speaker for the one in question. |
| **No** | LOCATE the source of the noise and REPAIR as needed. |

## Pinpoint Test E: No Sound From All Speakers

### Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

**Normal Operation and Fault Conditions**

Refer to ACM in Information and Entertainment System.

When the ignition is in the START position, the ACM receives voltage on the START signal circuit and mutes the speaker output. This prevents voltage spikes from producing popping noises through the speakers. A short to voltage on this circuit causes the ACM output to be muted continuously.

A short to ground or short to voltage in the circuitry to one of the speakers may cause multiple speakers to lose sound due to the built-in overload protection feature of the ACM . In this case, a speaker fault DTC sets.

**Possible Sources**

- Wiring, terminals, or connectors
- ACM

**PINPOINT TEST E : NO SOUND FROM ALL SPEAKERS**

**E1 CHECK FOR SPEAKER DTCS**

- Ignition ON.
- Using a diagnostic scan tool, perform the ACM self-test.

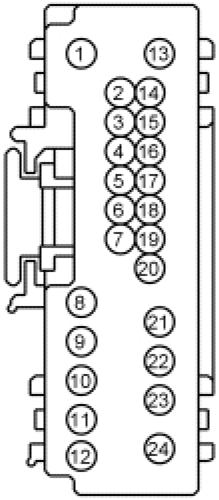
**Are any speaker fault Diagnostic Trouble Codes (DTCs) present?**

| | |
|---|---|
| **Yes** | REFER to the ACM DTC chart in Information and Entertainment System. |
| **No** | GO to E2. |

**E2 CHECK THE START SENSE CIRCUIT FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: ACM C290A .
- Ignition ON.
- Measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C290A Pin 15 | CDC54 (WH/GN) | — | Ground |

- 


**Is the voltage 5V or greater?**

| | |
|---|---|
| **Yes** | REPAIR the circuit. TEST the system for normal operation. |
| **No** | GO to E3. |

**E3 CHECK FOR CORRECT ACM OPERATION**

- Ignition OFF.
- Disconnect and inspect all the ACM connectors.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the ACM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test F: The LF Door Speaker Is Inoperative**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

**Normal Operation and Fault Conditions**

Refer to ACM in Information and Entertainment System.

| Root DTC Description | Failure Types | Fault Trigger Conditions |
|---|---|---|
| B1A01:01 | Speaker #1: General Electrical Failure | Set by the ACM when a failure is detected on the LF door or tweeter speaker circuits. |
| B1A01:11 | Speaker #1: Circuit Short to Ground | Set by the ACM when a short to ground is detected on the LF door or tweeter speaker circuits. When DTC B1A01:11 is set, the speaker output is disabled. |
| B1A01:12 | Speaker #1: Circuit Short to Battery | Set by the ACM when a short to battery is detected on the LF door or tweeter speaker circuits. When DTC B1A01:12 is set, the speaker output is disabled. |
| B1A01:13 | Speaker #1: Circuit Open | Set by the ACM when an open is detected on the LF door or tweeter speaker circuits. |

**Possible Sources**

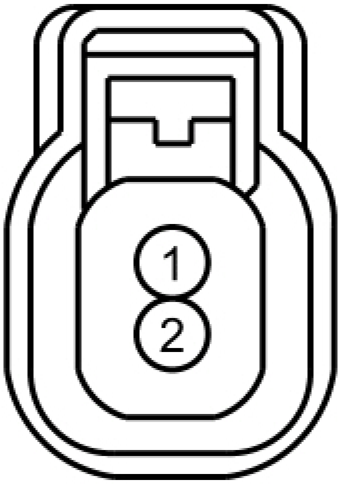
- Wiring, terminals, or connectors
- Speaker
- ACM

**PINPOINT TEST F : THE LF DOOR SPEAKER IS INOPERATIVE**

**F1 CHECK THE AUDIO SIGNAL TO THE LF DOOR SPEAKER**

- Ignition OFF.
- Disconnect: LF Door Speaker C523.
- Operate the audio system in radio tuner AM/FM mode.
- Measure the **AC voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C523 Pin 1 | VME07 (WH) | C523 Pin 2 | RME07 (WH/BN) |

- 

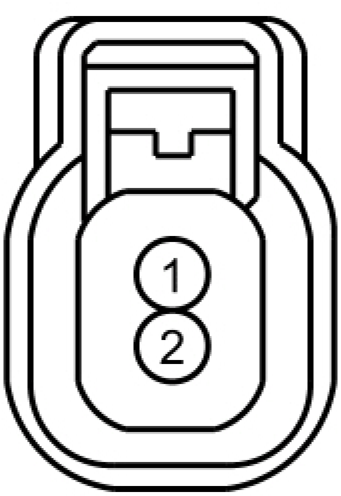

**Is a fluctuating AC voltage present?**

| | |
|---|---|
| **Yes** | INSTALL a new LF door speaker. REFER to Speaker — Door. |
| **No** | GO to F2. |

| |
|---|
| **F2 CHECK THE AUDIO CIRCUITS TO THE LF DOOR SPEAKER FOR A SHORT TO VOLTAGE** |

- Ignition OFF.
- Disconnect: ACM C290A .
- Ignition ON.
- Measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C523 Pin 1 | VME07 (WH) | — | Ground |
| C523 Pin 2 | RME07 (WH/BN) | — | Ground |

- 

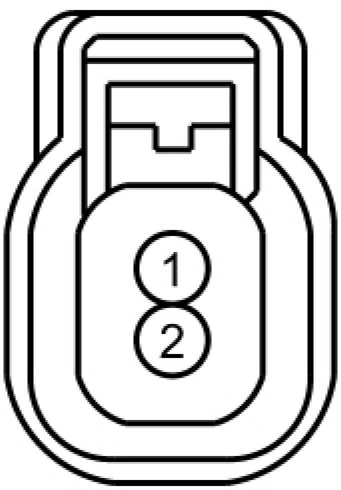
**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit in question. |
| **No** | GO to F3. |

## F3 CHECK THE AUDIO CIRCUITS TO THE LF DOOR SPEAKER FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: LF Tweeter Speaker C230.
- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C523 Pin 1 | VME07 (WH) | — | Ground |
| C523 Pin 2 | RME07 (WH/BN) | — | Ground |

- 


**Are the resistances greater than 10,000 ohms?**

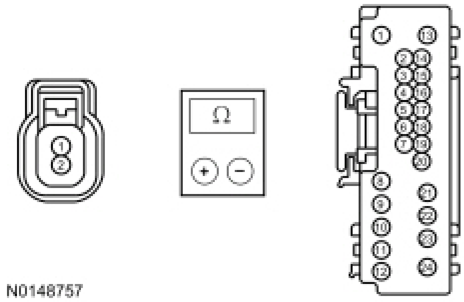
| | |
|---|---|
| **Yes** | GO to F4. |
| **No** | REPAIR the circuit in question. |

## F4 CHECK THE AUDIO CIRCUITS TO THE LF DOOR SPEAKER FOR AN OPEN

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C523 Pin 1 | VME07 (WH) | C290A Pin 8 | VME07 (WH) |

| | | | |
|---|---|---|---|
| C523 Pin 2 | RME07 (WH/BN) | C290A Pin 21 | RME07 (WH/BN) |

- 

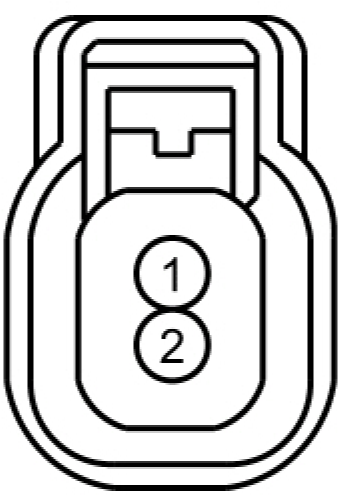

**Are the resistances less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to F5. |
| **No** | REPAIR the circuit in question. |

## F5 CHECK THE AUDIO CIRCUITS TO THE LF DOOR SPEAKER FOR A SHORT TOGETHER

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C523 Pin 1 | VME07 (WH) | C523 Pin 2 | RME07 (WH/BN) |

- 

**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to F6. |
| **No** | REPAIR the circuit in question. |

| F6 CHECK FOR CORRECT ACM OPERATION |
|---|
| • Ignition OFF.<br>• Disconnect and inspect all the ACM connectors.<br>• Repair:<br>  ▪ Corrosion (clean module pins or install new connectors or terminals)<br>  ▪ Damaged or bent pins (install new terminals or pins)<br>  ▪ Pushed-out pins (install new pins as necessary)<br>• Reconnect the ACM connectors. Make sure they seat and latch correctly.<br>• Operate the system and determine if the concern is still present.<br><br>**Is the concern still present?** |

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

### Pinpoint Test G: The LF Tweeter Speaker Is Inoperative

#### Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

#### Normal Operation and Fault Conditions

Refer to ACM in Information and Entertainment System.

| Root DTC Description | Failure Types | Fault Trigger Conditions |
|---|---|---|
| B1A01:01 | Speaker #1: General Electrical Failure | Set by the ACM when a failure is detected on the LF door or tweeter speaker circuits. |
| B1A01:11 | Speaker #1: Circuit Short to Ground | Set by the ACM when a short to ground is detected on the LF door or tweeter speaker circuits. When DTC B1A01:11 is set, the speaker output is disabled. |
| B1A01:12 | Speaker #1: Circuit Short to Battery | Set by the ACM when a short to battery is detected on the LF door or tweeter speaker circuits. When DTC B1A01:12 is set, the speaker output is disabled. |
| B1A01:13 | Speaker #1: Circuit Open | Set by the ACM when an open is detected on the LF door or tweeter speaker circuits. |

**Possible Sources**

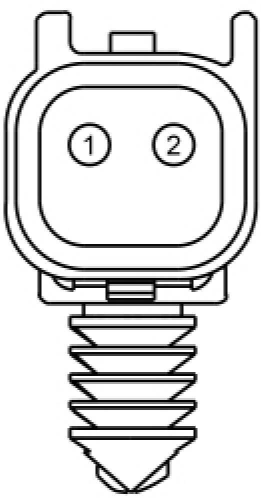
- Wiring, terminals, or connectors
- Speaker
- ACM

**PINPOINT TEST G : THE LF TWEETER SPEAKER IS INOPERATIVE**

**G1 CHECK THE AUDIO SIGNAL TO THE LF TWEETER SPEAKER**

- Ignition OFF.
- Disconnect: LF Tweeter Speaker C230.
- Operate the audio system in radio tuner AM/FM mode.
- Measure the **AC voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C230 Pin 1 | VME07 (WH) | C230 Pin 2 | RME07 (WH/BN) |

- 

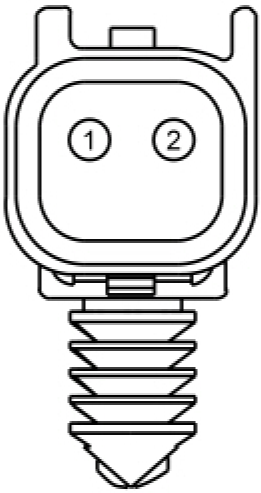

**Is a fluctuating AC voltage present?**

| | |
|---|---|
| **Yes** | INSTALL a new LF tweeter speaker. REFER to Speaker — A-Pillar. |
| **No** | GO to G2. |

**G2 CHECK THE AUDIO CIRCUITS TO THE LF TWEETER SPEAKER FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: ACM C290A .
- Ignition ON.
- Measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C230 Pin 1 | VME07 (WH) | — | Ground |
| C230 Pin 2 | RME07 (WH/BN) | — | Ground |

- 



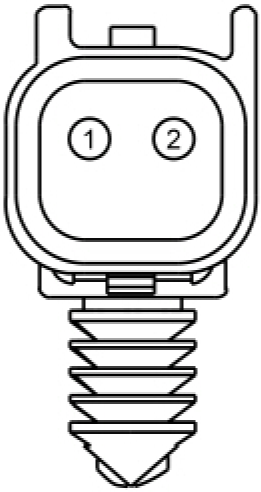
**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit in question. |
| **No** | GO to G3. |

## G3 CHECK THE AUDIO CIRCUITS TO THE LF TWEETER SPEAKER FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: LF Door Speaker C523.
- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C230 Pin 1 | VME07 (WH) | — | Ground |
| C230 Pin 2 | RME07 (WH/BN) | — | Ground |

- 

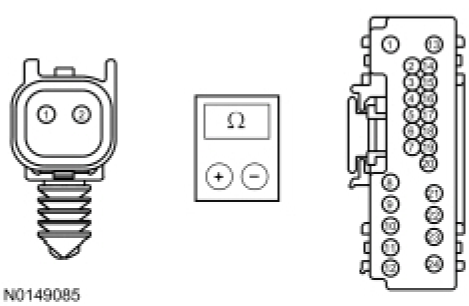
**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to G4. |
| **No** | REPAIR the circuit in question. |

**G4 CHECK THE AUDIO CIRCUITS TO THE LF TWEETER SPEAKER FOR AN OPEN**

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C230 Pin 1 | VME07 (WH) | C290A Pin 8 | VME07 (WH) |
| C230 Pin 2 | RME07 (WH/BN) | C290A Pin 21 | RME07 (WH/BN) |

- 

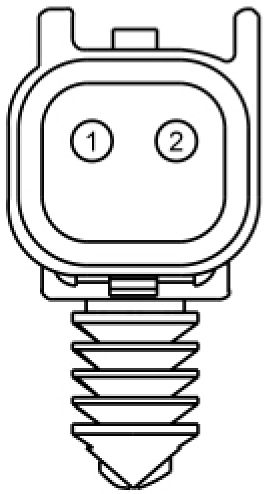

**Are the resistances less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to G5. |
| **No** | REPAIR the circuit in question. |

**G5 CHECK THE AUDIO CIRCUITS TO THE LF TWEETER SPEAKER FOR A SHORT TOGETHER**

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C230 Pin 1 | VME07 (WH) | C230 Pin 2 | RME07 (WH/BN) |

- 


**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to G6. |
| **No** | REPAIR the circuit in question. |

**G6 CHECK FOR CORRECT ACM OPERATION**

- Ignition OFF.
- Disconnect and inspect all the ACM connectors.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the ACM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test H: The RF Door Speaker Is Inoperative**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

**Normal Operation and Fault Conditions**

Refer to ACM in Information and Entertainment System.

| Root DTC Description | Failure Types | Fault Trigger Conditions |
|---|---|---|
| B1A02:01 | Speaker #2: General Electrical Failure | Set by the ACM when a failure is detected on the RF door or tweeter speaker circuits. |
| B1A02:11 | Speaker #2: Circuit Short to Ground | Set by the ACM when a short to ground is detected on the RF door or tweeter speaker circuits. When DTC B1A02:11 is set, the speaker output is disabled. |
| B1A02:12 | Speaker #2: Circuit Short to Battery | Set by the ACM when a short to battery is detected on the RF door or tweeter speaker circuits. When DTC B1A02:12 is set, the speaker output is disabled. |
| B1A02:13 | Speaker #2: Circuit Open | Set by the ACM when an open is detected on the RF door or tweeter speaker circuits. |

**Possible Sources**

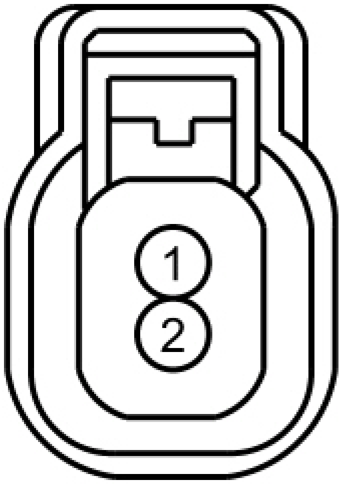
- Wiring, terminals, or connectors
- Speaker
- ACM

**PINPOINT TEST H : THE RF DOOR SPEAKER IS INOPERATIVE**

**H1 CHECK THE AUDIO SIGNAL TO THE RF DOOR SPEAKER**

- Ignition OFF.
- Disconnect: RF Door Speaker C612.
- Operate the audio system in radio tuner AM/FM mode.
- Measure the **AC voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C612 Pin 1 | VME10 (WH/VT) | C612 Pin 2 | RME10 (WH/OG) |

- 

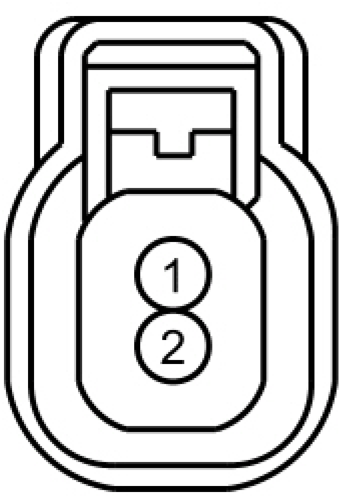

**Is a fluctuating AC voltage present?**

| | |
|---|---|
| **Yes** | INSTALL a new RF door speaker. REFER to Speaker — Door. |
| **No** | GO to H2. |

| **H2 CHECK THE AUDIO CIRCUITS TO THE RF DOOR SPEAKER FOR A SHORT TO VOLTAGE** |
|---|

- Ignition OFF.
- Disconnect: ACM C290A .
- Ignition ON.
- Measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C612 Pin 1 | VME10 (WH/VT) | — | Ground |
| C612 Pin 2 | RME10 (WH/OG) | — | Ground |

- 

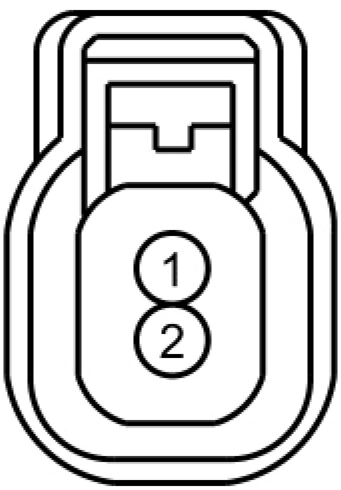
**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit in question. |
| **No** | GO to H3. |

**H3 CHECK THE AUDIO CIRCUITS TO THE RF DOOR SPEAKER FOR A SHORT TO GROUND**

- Ignition OFF.
- Disconnect: RF Door Speaker C231.
- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C612 Pin 1 | VME10 (WH/VT) | — | Ground |
| C612 Pin 2 | RME10 (WH/OG) | — | Ground |

- 


**Are the resistances greater than 10,000 ohms?**

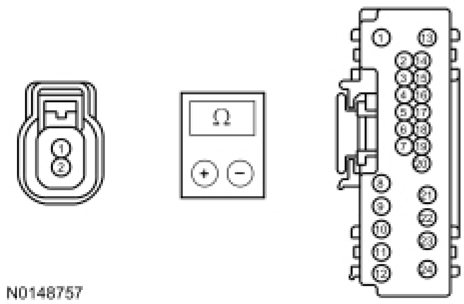
| | |
|---|---|
| **Yes** | GO to H4. |
| **No** | REPAIR the circuit in question. |

**H4 CHECK THE AUDIO CIRCUITS TO THE RF DOOR SPEAKER FOR AN OPEN**

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C612 Pin 1 | VME10 (WH/VT) | C290A Pin 11 | VME10 (WH/VT) |

| | | | |
|---|---|---|---|
| C612 Pin 2 | RME10 (WH/OG) | C290A Pin 12 | RME10 (WH/OG) |

- 

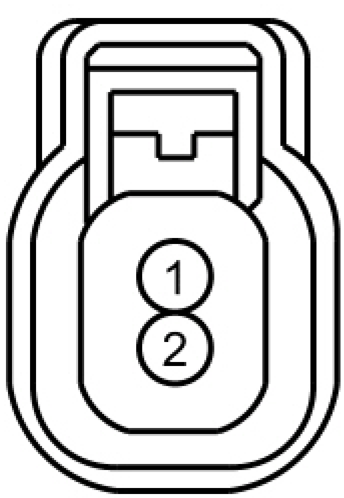

**Are the resistances less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to H5. |
| **No** | REPAIR the circuit in question. |

**H5 CHECK THE AUDIO CIRCUITS TO THE RF DOOR SPEAKER FOR A SHORT TOGETHER**

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C612 Pin 1 | VME10 (WH/VT) | C612 Pin 2 | RME10 (WH/OG) |

- 

**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to H6. |
| **No** | REPAIR the circuit in question. |

| H6 CHECK FOR CORRECT ACM OPERATION |
|---|
| • Ignition OFF.<br>• Disconnect and inspect all the ACM connectors.<br>• Repair:<br>  ▪ Corrosion (clean module pins or install new connectors or terminals)<br>  ▪ Damaged or bent pins (install new terminals or pins)<br>  ▪ Pushed-out pins (install new pins as necessary)<br>• Reconnect the ACM connectors. Make sure they seat and latch correctly.<br>• Operate the system and determine if the concern is still present.<br><br>**Is the concern still present?** |

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test I: The RF Tweeter Speaker Is Inoperative**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

**Normal Operation and Fault Conditions**

Refer to ACM in Information and Entertainment System.

| Root DTC Description | Failure Types | Fault Trigger Conditions |
|---|---|---|
| B1A02:01 | Speaker #2: General Electrical Failure | Set by the ACM when a failure is detected on the RF door or tweeter speaker circuits. |
| B1A02:11 | Speaker #2: Circuit Short to Ground | Set by the ACM when a short to ground is detected on the RF door or tweeter speaker circuits. When DTC B1A02:11 is set, the speaker output is disabled. |
| B1A02:12 | Speaker #2: Circuit Short to Battery | Set by the ACM when a short to battery is detected on the RF door or tweeter speaker circuits. When DTC B1A02:12 is set, the speaker output is disabled. |
| B1A02:13 | Speaker #2: Circuit Open | Set by the ACM when an open is detected on the RF door or tweeter speaker circuits. |

**Possible Sources**

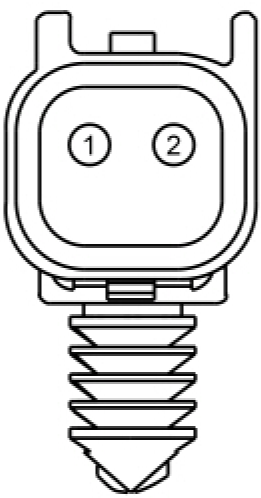
- Wiring, terminals, or connectors
- Speaker
- ACM

**PINPOINT TEST I : THE RF TWEETER SPEAKER IS INOPERATIVE**

**I1 CHECK THE AUDIO SIGNAL TO THE RF TWEETER SPEAKER**

- Ignition OFF.
- Disconnect: RF Tweeter Speaker C231.
- Operate the audio system in radio tuner AM/FM mode.
- Measure the **AC voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C231 Pin 1 | VME10 (WH/VT) | C231 Pin 2 | RME10 (WH/OG) |

- 

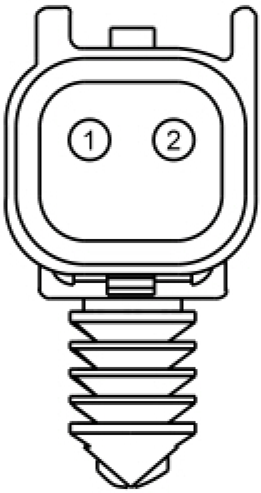

**Is a fluctuating AC voltage present?**

| | |
|---|---|
| **Yes** | INSTALL a new RF tweeter speaker. REFER to Speaker — A-Pillar. |
| **No** | GO to I2. |

**I2 CHECK THE AUDIO CIRCUITS TO THE RF TWEETER SPEAKER FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: ACM C290A .
- Ignition ON.
- Measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C231 Pin 1 | VME10 (WH/VT) | — | Ground |
| C231 Pin 2 | RME10 (WH/OG) | — | Ground |

- 

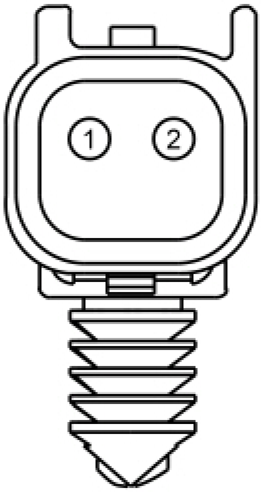

**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit in question. |
| **No** | GO to I3. |

**I3 CHECK THE AUDIO CIRCUITS TO THE RF TWEETER SPEAKER FOR A SHORT TO GROUND**

- Ignition OFF.
- Disconnect: RF Door Speaker C612.
- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C231 Pin 1 | VME10 (WH/VT) | — | Ground |
| C231 Pin 2 | RME10 (WH/OG) | — | Ground |

- 

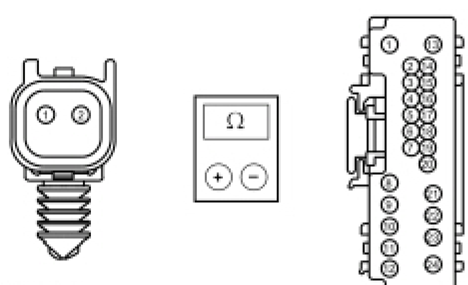
**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to I4. |
| **No** | REPAIR the circuit in question. |

| I4 CHECK THE AUDIO CIRCUITS TO THE RF TWEETER SPEAKER FOR AN OPEN |
|---|

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C231 Pin 1 | VME10 (WH/VT) | C290A Pin 11 | VME10 (WH/VT) |
| C231 Pin 2 | RME10 (WH/OG) | C290A Pin 12 | RME10 (WH/OG) |

- 

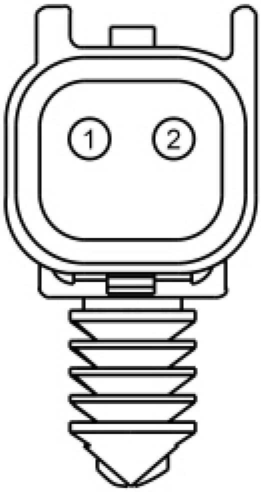

**Are the resistances less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to I5. |
| **No** | REPAIR the circuit in question. |

| I5 CHECK THE AUDIO CIRCUITS TO THE RF TWEETER SPEAKER FOR A SHORT TOGETHER |
|---|

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C231 Pin 1 | VME10 (WH/VT) | C231 Pin 2 | RME10 (WH/OG) |

- 


**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to I6. |
| **No** | REPAIR the circuit in question. |

**I6 CHECK FOR CORRECT ACM OPERATION**

- Ignition OFF.
- Disconnect and inspect all the ACM connectors.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the ACM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBsTSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test J: The RR Door Speaker Is Inoperative**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

**Normal Operation and Fault Conditions**

Refer to ACM in Information and Entertainment System.

| Root DTC Description | Failure Types | Fault Trigger Conditions |
|---|---|---|
| B1A03:01 | Speaker #3: General Electrical Failure | Set by the ACM when a failure is detected on the RR door speaker circuits. |
| B1A03:11 | Speaker #3: Circuit Short to Ground | Set by the ACM when a short to ground is detected on the RR door speaker circuits. When DTC B1A03:11 is set, the speaker output is disabled. |
| B1A03:12 | Speaker #3: Circuit Short to Battery | Set by the ACM when a short to battery is detected on the RR door speaker circuits. When DTC B1A03:12 is set, the speaker output is disabled. |
| B1A03:13 | Speaker #3: Circuit Open | Set by the ACM when an open is detected on the RR door speaker circuits. |

**Possible Sources**

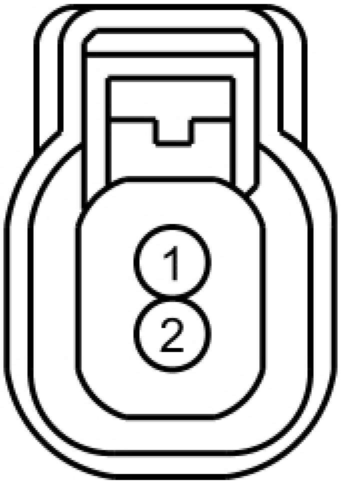
- Wiring, terminals, or connectors
- Speaker
- ACM

**PINPOINT TEST J : THE RR DOOR SPEAKER IS INOPERATIVE**

**J1 CHECK THE AUDIO SIGNAL TO THE RR DOOR SPEAKER**

- Ignition OFF.
- Disconnect: RR Door Speaker C802.
- Operate the audio system in radio tuner AM/FM mode.
- Measure the **AC voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C802 Pin 1 | VME12 (BN/WH) | C802 Pin 2 | RME12 (BN/BU) |

- 

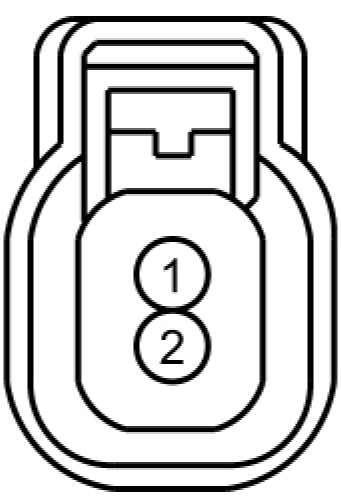

**Is a fluctuating AC voltage present?**

| | |
|---|---|
| **Yes** | INSTALL a new RR door speaker. REFER to Speaker — Door. |
| **No** | GO to J2. |

**J2 CHECK THE AUDIO CIRCUITS TO THE RR DOOR SPEAKER FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: ACM C290A .
- Ignition ON.
- Measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C802 Pin 1 | VME12 (BN/WH) | — | Ground |
| C802 Pin 2 | RME12 (BN/BU) | — | Ground |

- 

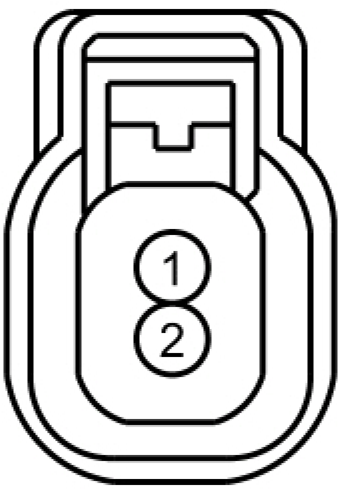
**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit in question. |
| **No** | GO to J3. |

## J3 CHECK THE AUDIO CIRCUITS TO THE RR DOOR SPEAKER FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C802 Pin 1 | VME12 (BN/WH) | — | Ground |
| C802 Pin 2 | RME12 (BN/BU) | — | Ground |

- 

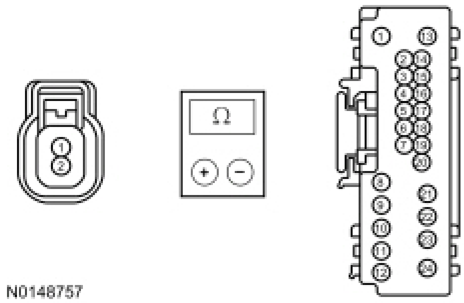

**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to J4. |
| **No** | REPAIR the circuit in question. |

## J4 CHECK THE AUDIO CIRCUITS TO THE RR DOOR SPEAKER FOR AN OPEN

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C802 Pin 1 | VME12 (BN/WH) | C290A Pin 10 | VME12 (BN/WH) |
| C802 Pin 2 | RME12 (BN/BU) | C290A Pin 23 | RME12 (BN/BU) |

- 

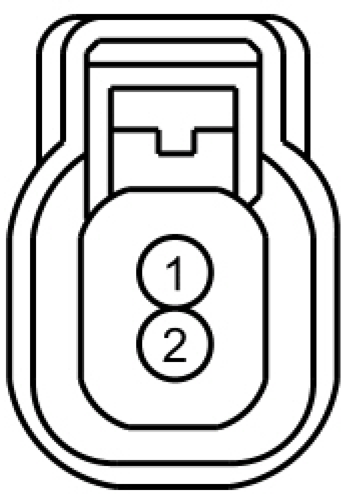

**Are the resistances less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to J5. |
| **No** | REPAIR the circuit in question. |

**J5 CHECK THE AUDIO CIRCUITS TO THE RR DOOR SPEAKER FOR A SHORT TOGETHER**

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C802 Pin 1 | VME12 (BN/WH) | C802 Pin 2 | RME12 (BN/BU) |

- 


**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to J6. |
| **No** | REPAIR the circuit in question. |

**J6 CHECK FOR CORRECT ACM OPERATION**

- Ignition OFF.
- Disconnect and inspect all the ACM connectors.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the ACM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test K: The LR Door Speaker Is Inoperative**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

**Normal Operation and Fault Conditions**

Refer to ACM in Information and Entertainment System.

| Root DTC Description | Failure Types | Fault Trigger Conditions |
|---|---|---|
| B1A04:01 | Speaker #4: General Electrical Failure | Set by the ACM when a failure is detected on the LR door speaker circuits. |
| B1A04:11 | Speaker #4: Circuit Short to Ground | Set by the ACM when a short to ground is detected on the LR door speaker circuits. When DTC B1A04:11 is set, the speaker output is disabled. |
| B1A04:12 | Speaker #4: Circuit Short to Battery | Set by the ACM when a short to battery is detected on the LR door speaker circuits. When DTC B1A04:12 is set, the speaker output is disabled. |
| B1A04:13 | Speaker #4: Circuit Open | Set by the ACM when an open is detected on the LR door speaker circuits. |

**Possible Sources**

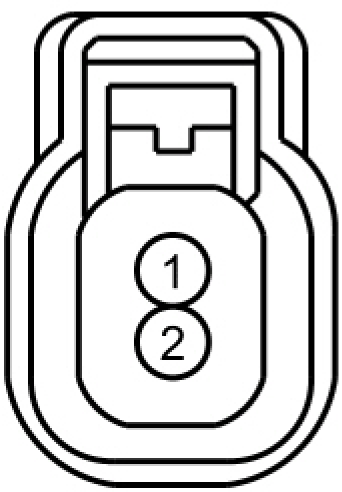
- Wiring, terminals, or connectors
- Speaker
- ACM

**PINPOINT TEST K : THE LR DOOR SPEAKER IS INOPERATIVE**

**K1 CHECK THE AUDIO SIGNAL TO THE LR DOOR SPEAKER**

- Ignition OFF.
- Disconnect: LR Door Speaker C702.
- Operate the audio system in radio tuner AM/FM mode.
- Measure the **AC voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C702 Pin 1 | VME09 (WH/GN) | C702 Pin 2 | RME09 (BN/YE) |

- 


**Is a fluctuating AC voltage present?**

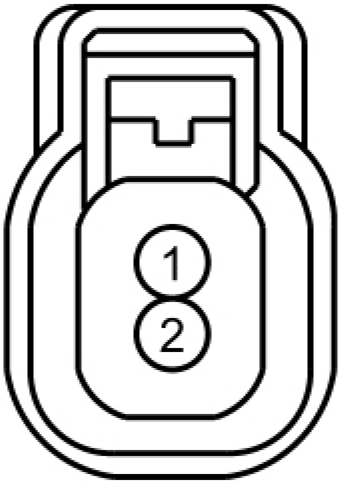
| | |
|---|---|
| **Yes** | INSTALL a new LR door speaker. REFER to Speaker — Door. |
| **No** | GO to K2. |

**K2 CHECK THE AUDIO CIRCUITS TO THE LR DOOR SPEAKER FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: ACM C290A .
- Ignition ON.
- Measure the **voltage** between:

| Positive Lead | Negative Lead |
|---|---|

| Pin | Circuit | Pin | Circuit |
|---|---|---|---|
| C702 Pin 1 | VME09 (WH/GN) | — | Ground |
| C702 Pin 2 | RME09 (BN/YE) | — | Ground |

- 

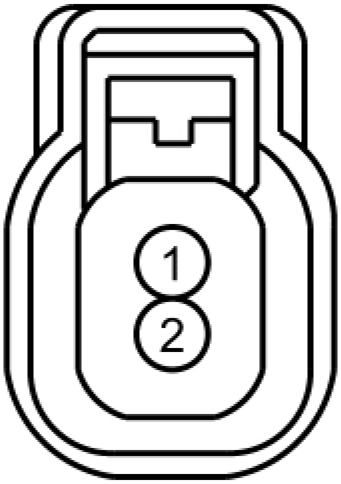

**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit in question. |
| **No** | GO to K3. |

**K3 CHECK THE AUDIO CIRCUITS TO THE LR DOOR SPEAKER FOR A SHORT TO GROUND**

- Ignition OFF.
- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C702 Pin 1 | VME09 (WH/GN) | — | Ground |
| C702 Pin 2 | RME09 (BN/YE) | — | Ground |

- 

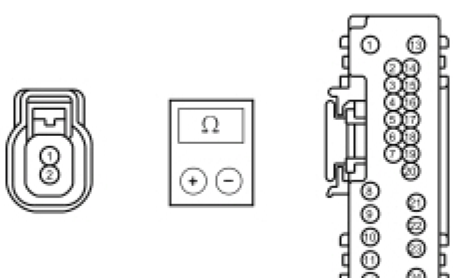

**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to K4. |
| **No** | REPAIR the circuit in question. |

## K4 CHECK THE AUDIO CIRCUITS TO THE LR DOOR SPEAKER FOR AN OPEN

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C702 Pin 1 | VME09 (WH/GN) | C290A Pin 9 | VME09 (WH/GN) |
| C702 Pin 2 | RME09 (BN/YE) | C290A Pin 22 | RME09 (BN/YE) |

- 
N0148757

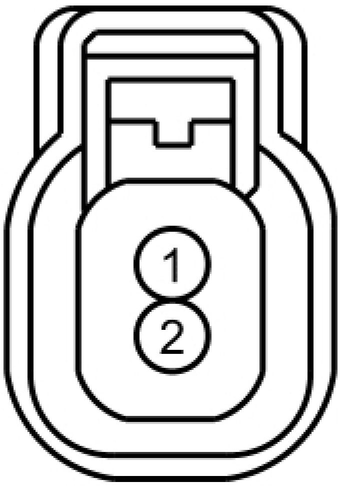
**Are the resistances less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to K5. |
| **No** | REPAIR the circuit in question. |

## K5 CHECK THE AUDIO CIRCUITS TO THE LR DOOR SPEAKER FOR A SHORT TOGETHER

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C702 Pin 1 | VME09 (WH/GN) | C702 Pin 2 | RME09 (BN/YE) |

- 


**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to K6. |
| **No** | REPAIR the circuit in question. |

**K6 CHECK FOR CORRECT ACM OPERATION**

- Ignition OFF.
- Disconnect and inspect all the ACM connectors.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the ACM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

## Pinpoint Test L: No Sound From All SYNC® Audio Sources (Bluetooth, USB , Audio Input Jack)

### Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

### Normal Operation and Fault Conditions

Refer to SYNC® System in Information and Entertainment System.

The APIM transmits left and right channel analog audio signals to the ACM . These signals include inputs from the USB port, audio input jack, and Bluetooth media device. If a USB mass storage device is used to play an audio file, the SYNC® system only plays audio files that do not have DRM protection.

### Possible Sources

- Customer's device
- Wiring, terminals, or connectors
- APIM
- ACM

### PINPOINT TEST L : NO SOUND FROM ALL SYNC® AUDIO SOURCES (BLUETOOTH, USB , AUDIO INPUT JACK)

#### L1 VERIFY THE OPERATION OF THE SYNC® SYSTEM AUDIO SOURCES

- Connect: Multi-Media Interface Tester 105-00120.
- Using the Multi-Media Interface Tester 105-00120, test the audio output for the audio input jack, USB port, and Bluetooth.

**Is there poor quality, distorted, or no sound from all SYNC® audio sources?**

| | |
|---|---|
| **Yes** | GO to L2. |
| **No** | If there is a concern with one or more, but not all audio sources, GO to Symptom Chart — SYNC® System to diagnose the observed symptom.<br>If all audio sources operate correctly, the concern is with the customer's device. |

#### L2 CHECK THE AUDIO SIGNALS TO THE ACM

**NOTICE: This pinpoint test step directs testing circuits using a back-probe method. Use the special back-probe tool POM6411. Do not force test leads or other probes into connectors. Adequate care must be exercised to avoid connector terminal damage, while ensuring that good electrical contact is made with the circuit or terminal. Failure to follow these instructions may cause damage to wiring, terminals, or connectors, and may cause subsequent electrical faults.**

- Operate the audio system in SYNC® mode and play an audio file.
- Measure the **AC voltage** by back-probing ACM connector C290B:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |

| | | | |
|---|---|---|---|
| C290B Pin 1 | VME52 (BU) | C290B Pin 2 | RME52 (GY/OG) |
| C290B Pin 9 | VME53 (VT/GN) | C290B Pin 10 | RME53 (BN/WH) |

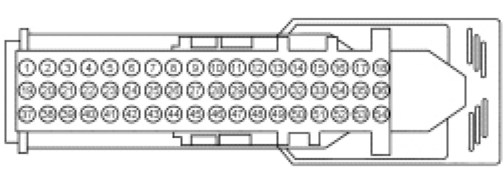
**Is a fluctuating AC voltage present?**

| | |
|---|---|
| **Yes** | GO to L7. |
| **No** | GO to L3. |

**L3 CHECK THE AUDIO CIRCUITS TO THE ACM FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: ACM C290B .
- Disconnect: APIM C2383 .
- Ignition ON.
- Measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C2383 Pin 23 | VME53 (VT/GN) | — | Ground |
| C2383 Pin 24 | RME53 (BN/WH) | — | Ground |
| C2383 Pin 25 | VME52 (BU) | — | Ground |
| C2383 Pin 26 | RME52 (GY/OG) | — | Ground |



- 

**Is any voltage present?**

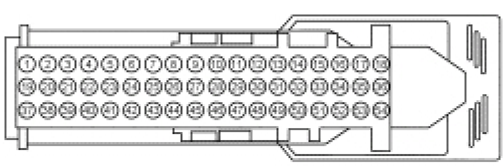
| | |
|---|---|
| **Yes** | REPAIR the circuit in question. |

| | |
|---|---|
| **No** | GO to L4. |

## L4 CHECK THE AUDIO CIRCUITS TO THE ACM FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C2383 Pin 23 | VME53 (VT/GN) | — | Ground |
| C2383 Pin 24 | RME53 (BN/WH) | — | Ground |
| C2383 Pin 25 | VME52 (BU) | — | Ground |
| C2383 Pin 26 | RME52 (GY/OG) | — | Ground |



- 

**Are the resistances greater than 10,000 ohms?**

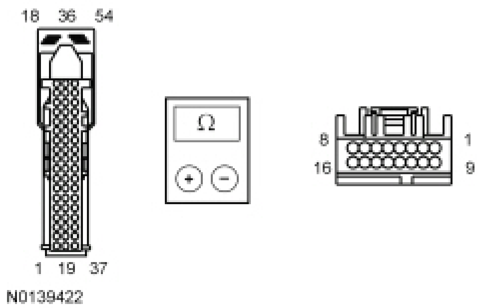
| | |
|---|---|
| **Yes** | GO to L5. |
| **No** | REPAIR the circuit in question. |

## L5 CHECK THE CIRCUITS TO THE ACM FOR AN OPEN

- Ignition OFF.
- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C2383 Pin 23 | VME53 (VT/GN) | C290B Pin 9 | VME53 (VT/GN) |
| C2383 Pin 24 | RME53 (BN/WH) | C290B Pin 10 | RME53 (BN/WH) |

| | | | |
|---|---|---|---|
| C2383 Pin 25 | VME52 (BU) | C290B Pin 1 | VME52 (BU) |
| C2383 Pin 26 | RME52 (GY/OG) | C290B Pin 2 | RME52 (GY/OG) |

- 


**Are the resistances less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to L6. |
| **No** | REPAIR the circuit in question. |

**L6 CHECK FOR CORRECT APIM OPERATION**

- Ignition OFF.
- Disconnect and inspect the APIM connector.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the APIM connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the APIM to re-initialize.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00. |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**L7 CHECK FOR CORRECT ACM OPERATION**

- Ignition OFF.
- Disconnect and inspect all the ACM connectors.
- Repair:

- Corrosion (clean module pins or install new connectors or terminals)
- Damaged or bent pins (install new terminals or pins)
- Pushed-out pins (install new pins as necessary)

- Reconnect the ACM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test M: The SYNC® System Is Inoperative (No Response Is Received From Phone, Voice, And Media Inputs)**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

**Normal Operation and Fault Conditions**

Refer to SYNC® System in Information and Entertainment System.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| U3000:04 | Control Module: System Internal Failure | Set by the APIM when it detects a fault due to a device conflict or internal failure. |

**Possible Sources**

- Customer's device
- APIM

**PINPOINT TEST M : THE SYNC® SYSTEM IS INOPERATIVE (NO RESPONSE IS RECEIVED FROM PHONE, VOICE, AND MEDIA INPUTS)**

**M1 VERIFY THAT THE APIM PASSES THE NETWORK TEST**

- Using a diagnostic scan tool, perform the network test.

**Did the APIM pass the network test?**

| | |
|---|---|
| **Yes** | GO to M2. |
| **No** | REFER to Section 418-00. |

**M2 CHECK THE SYNC® SYSTEM INPUTS FOR FUNCTIONALITY**

**NOTE:** *Carrying out a Master Reset returns all preference settings to the factory defaults, erases the phone book and call histories, and deletes any devices paired with the SYNC® system.*

- Connect: Multi-Media Interface Tester 105-00120.
- Using the Multi-Media Interface Tester 105-00120, test the audio output for the audio input jack, USB port, and Bluetooth connection using the VOICE switch on the steering wheel controls to enter each mode.

**Do all of the SYNC® inputs function correctly?**

| | |
|---|---|
| **Yes** | The SYNC® system is operating correctly at this time. CARRY OUT a SYNC® system Master Reset. REFER to the Owner's Literature.<br>REVIEW the SYNC® system operation with the customer. If the customer's device still does not operate correctly, the fault is with the customer's device. |
| **No** | GO to M3. |

**M3 RESET THE APIM AND RECHECK SYNC® SYSTEM OPERATION**

**NOTE:** *Carrying out a Master Reset returns all preference settings to the factory defaults, erases the phone book and call histories, and deletes any devices paired with the SYNC® system.*

- Carry out an APIM power reset by disconnecting the battery for 1 minute, then reconnecting it.
- Connect: Multi-Media Interface Tester 105-00120.
- Using the Multi-Media Interface Tester 105-00120, test the audio output for the audio input jack, USB port, and Bluetooth connection using the VOICE switch on the steering wheel controls to enter each mode.

**Do all of the SYNC® inputs function correctly?**

| | |
|---|---|
| **Yes** | The SYNC® system is operating correctly at this time. CARRY OUT a SYNC® system Master Reset. REFER to the Owner's Literature.<br>REVIEW the SYNC® system operation with the customer. If the customer's device still does not operate correctly, the fault is with the customer's device. |
| **No** | If only some, but not all of the inputs are inoperative, GO to Symptom Chart — SYNC® System to diagnose the observed symptom.<br>If all inputs are inoperative and/or DTC U3000:04 is present, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00. |

**Pinpoint Test N: The SYNC® System Voice Or Tone Prompts, TTS Feature, Or Ringtones Are Inoperative Or Do Not Operate Correctly, Or During A Phone Call, No Incoming Audio Is Heard In The Vehicle**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System for schematic and connector information.

**Normal Operation and Fault Conditions**

Refer to SYNC® System and Voice Recognition in Information and Entertainment System.

**Possible Sources**

- Customer's setting
- Wiring, terminals, or connectors
- APIM
- ACM

**PINPOINT TEST N : THE SYNC® SYSTEM VOICE OR TONE PROMPTS, TTS FEATURE, OR RINGTONES ARE INOPERATIVE OR DO NOT OPERATE CORRECTLY, OR DURING A PHONE CALL, NO INCOMING AUDIO IS HEARD IN THE VEHICLE**

**N1 CHECK THE AUDIBLE PROMPT SETTING**

- Operate the audio system in SYNC® mode.
- Verify the audible prompts are enabled. Refer to "SYNC® Voice Recognition Feature" in the Owner's Literature.
- Operate the VOICE button on the steering wheel controls and observe the SYNC® audible prompt.

**Does the SYNC® system produce an audible prompt correctly?**

| | |
|---|---|
| **Yes** | The system is operating correctly at this time. The concern was caused by the customer's setting. INSTRUCT the customer in the correct operation of the audible prompt feature. |
| **No** | GO to N2. |

**N2 CHECK FOR A VOLTAGE SIGNAL FROM THE APIM**

**NOTICE: This pinpoint test step directs testing circuits using a back-probe method. Use the special back-probe tool POM6411. Do not force test leads or other probes into connectors. Adequate care must be exercised to avoid connector terminal damage while ensuring that good electrical contact is made with the circuit or terminal. Failure to follow these instructions may cause damage to wiring, terminals, or connectors, and may cause subsequent electrical faults.**

- Operate the audio system in SYNC® mode.
- While operating the VOICE button repeatedly, measure the **AC voltage** by back-probing ACM connector C290B:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C290B Pin 4 | VMN14 (WH/VT) | C290B Pin 5 | RMN14 (GY/BN) |

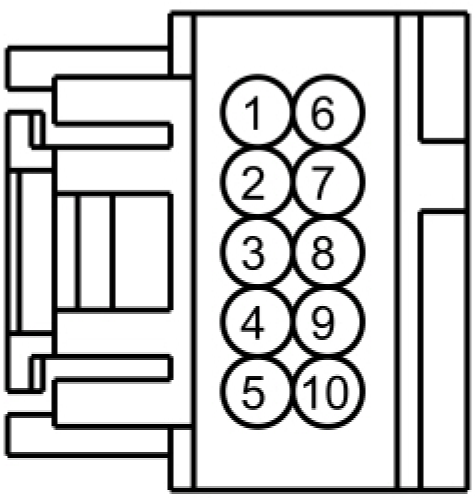
**Is a fluctuating AC voltage present each time the VOICE switch is operated?**

| | |
|---|---|
| **Yes** | GO to N7. |
| **No** | GO to N3. |

## N3 CHECK THE AUDIBLE PROMPT CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ACM C290B .
- Disconnect: APIM C2383 .
- Ignition ON.
- Measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C290B Pin 4 | VMN14 (WH/VT) | — | Ground |
| C290B Pin 5 | RMN14 (GY/BN) | — | Ground |

- 

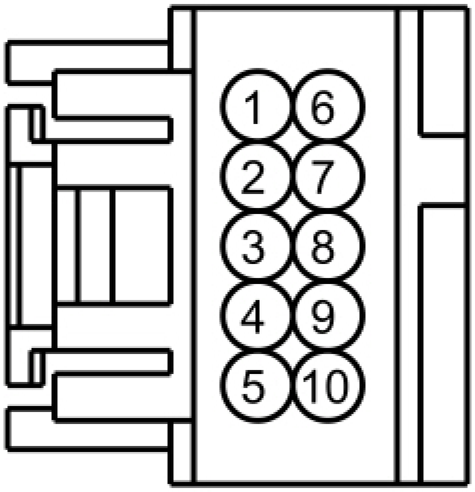

**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit in question. |
| **No** | GO to N4. |

## N4 CHECK THE AUDIBLE PROMPT CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C290B Pin 4 | VMN14 (WH/VT) | — | Ground |
| C290B Pin 5 | RMN14 (GY/BN) | — | Ground |

- 

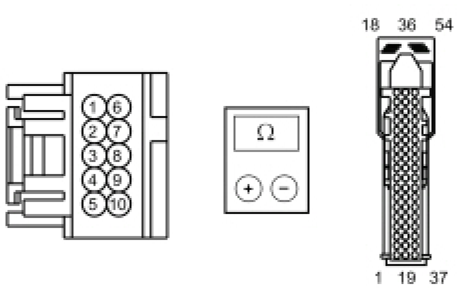

**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to N5. |
| **No** | REPAIR the circuit in question. |

**N5 CHECK THE AUDIBLE PROMPT CIRCUITS FOR AN OPEN**

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C290B Pin 4 | VMN14 (WH/VT) | C2383 Pin 3 | VMN14 (WH/VT) |
| C290B Pin 5 | RMN14 (GY/BN) | C2383 Pin 4 | RMN14 (GY/BN) |

- 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**Are the resistances less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to N6. |
| **No** | REPAIR the circuit in question. |

**N6 CHECK FOR CORRECT APIM OPERATION**

- Ignition OFF.
- Disconnect and inspect the APIM connector.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the APIM connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the APIM to re-initialize.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00. |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**N7 CHECK FOR CORRECT ACM OPERATION**

- Ignition OFF.
- Disconnect and inspect all the ACM connectors.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the ACM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

## Pinpoint Test O: All SYNC® Services Features Are Inoperative Or Inaccurate

### Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

### Normal Operation and Fault Conditions

Refer to SYNC® Services in Information and Entertainment System.

### Possible Sources

- Unregistered SYNC® owner account
- Unregistered or expired SYNC® Services subscription
- Unregistered mobile phone number
- Mobile phone being used not selected as "Active"
- SYNC® Services server
- APIM
- GPSM

**PINPOINT TEST O : ALL SYNC® SERVICES FEATURES ARE INOPERATIVE OR INACCURATE**

**O1 VERIFY THE CUSTOMER'S SYNC® OWNER ACCOUNT AND SYNC® SERVICES REGISTRATION AND SUBSCRIPTION**

- Verify that the customer has a SYNC® Owner Account on SyncMyRide
- Verify that the customer has registered for SYNC® Services and has an active subscription.

**Does the customer have a SYNC® Owner Account and an active SYNC® Services subscription?**

| | |
|---|---|
| **Yes** | GO to O2. |
| **No** | If the customer does not have a SYNC® Owner Account, INFORM the customer that an account must be created in order to sign up for SYNC® Services. REFER to SyncMyRide.<br>If the customer does not have an active SYNC® Services subscription, INFORM the customer that in order for SYNC® Services to function, the subscription needs to be activated. REFER to SyncMyRide. |

**O2 VERIFY THE CUSTOMER'S REGISTERED PHONE NUMBER AND STATUS**

- Verify that the mobile phone being used has the correct phone number registered with SYNC® Services.
- Verify that the mobile phone being used is selected as "Active" on SyncMyRide.

**Is the customer using the registered and active mobile phone associated with the SYNC® services account?**

| | |
|---|---|
| **Yes** | GO to O3. |
| **No** | INFORM the customer that the correct phone number must be registered on the SYNC® website and selected as "Active." REFER to SyncMyRide. |

## O3 CHECK FOR APIM DTCS

- Ignition ON.
- Using a diagnostic scan tool, perform the APIM self-test.

**Are any DTCs recorded in the APIM ?**

| | |
|---|---|
| **Yes** | REFER to the APIM DTC Chart in Information and Entertainment System. |
| **No** | GO to O4. |

## O4 CHECK FOR GPSM DTCS

- Using a diagnostic scan tool, perform the GPSM self-test.

**Are any DTCs recorded in the GPSM ?**

| | |
|---|---|
| **Yes** | REFER to the GPSM DTC Chart in Information and Entertainment System. |
| **No** | GO to O5. |

## O5 VERIFY THE CUSTOMER'S PHONE CONNECTION TO SYNC® SERVICES

- Advise the customer to connect their phone to SYNC® Services. Caller ID blocking must be disabled on the customer's phone. Refer to the Owner's Literature.
- Operate a SYNC® Services feature.
- Verify that the phone call is placed and that the phone is connected to SYNC® Services.

**Does the phone connect to SYNC® Services and operate correctly?**

| | |
|---|---|
| **Yes** | GO to O6. |
| **No** | GO to Pinpoint Test S. |

## O6 OBSERVE THE SYNC® SERVICES SUBSCRIPTION COMMERCIAL

- Once the customer's phone is connected to SYNC® Services, observe the subscription commercial.

**Does the SYNC® Services subscription commercial repeat with no selection option?**

| | |
|---|---|
| **Yes** | There is a potential SYNC® Services server issue, and the problem does not reside in the vehicle's audio system. CONTACT the Ford Technical Hotline for assistance. |
| **No** | GO to O7. |

## O7 CHECK THE OPERATION OF THE DIRECTIONS FROM SYNC® SERVICES

- When prompted, select directions from SYNC® Services and observe the audible prompts from the audio system.

**Is a single chime, audible prompt, then a short chime heard, indicating the navigation route is being sent to the vehicle?**

| | |
|---|---|
| **Yes** | SYNC® Services is operating correctly at this time. |
| **No** | If a single chime, audible prompt, then a repeating chime is heard, indicating the navigation route is being sent to the vehicle, GO to O8.<br>If no chime or audible prompt is heard, indicating that the system was unable to locate the vehicle, there is a potential SYNC® Services server issue. The problem does not reside in the vehicle's audio system. CONTACT the Ford Technical Hotline for assistance. |

**O8 CHECK FOR CORRECT APIM OPERATION**

- Ignition OFF.
- Disconnect and inspect the APIM connector.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the APIM connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the APIM to re-initialize.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00. |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

## Pinpoint Test P: Voice Recognition Is Inoperative Or Does Not Operate Correctly, Or During A Phone Call, Poor Quality Or No Outgoing Audio Is Heard On The Outside Device

### Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System for schematic and connector information.

### Normal Operation and Fault Conditions

Refer to Voice Recognition in Information and Entertainment System.

### DTC Fault Trigger Conditions

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| B1D79:01 | Microphone Input: General Electrical Failure | Set by the APIM when a microphone is not detected during the on-demand self-test. |
| B1D79:11 | Microphone Input: Circuit Short To Ground | Set by the APIM when short to ground is detected on the microphone circuits. |
| B1D79:12 | Microphone Input: Circuit Short To Battery | Set by the APIM when short to voltage is detected on the microphone circuits. |
| B1D79:13 | Microphone Input: Circuit Open | Set by the APIM when an open is detected on the microphone circuits. |

### Possible Sources

- Insufficiently quiet vehicle
- Inadequate phone reception
- Wiring, terminals, or connectors
- Microphone
- APIM

### PINPOINT TEST P : VOICE RECOGNITION IS INOPERATIVE OR DOES NOT OPERATE CORRECTLY, OR DURING A PHONE CALL, POOR QUALITY OR NO OUTGOING AUDIO IS HEARD ON THE OUTSIDE DEVICE

**P1 CHECK THE AUDIBLE PROMPT FUNCTIONALITY**

- Operate the audio system in radio tuner AM/FM mode.
- Operate the VOICE steering wheel switch and listen for the SYNC® audible prompt.

**Does the SYNC® system produce an audible prompt correctly, indicating voice recognition mode has been entered?**

| | |
|---|---|
| **Yes** | GO to P3. |
| **No** | GO to P2. |

**P2 CHECK THE OPERATION OF THE RH STEERING WHEEL SWITCHES**

- Operate the audio system in AM/FM mode.
- Operate the VOLUME +/-, SEEK +/-, MEDIA, PHONE, and OK steering wheel switches.

**Do the VOLUME +/-, SEEK +/-, MEDIA, PHONE, and OK steering wheel switches operate correctly?**

| | |
|---|---|
| **Yes** | GO to Pinpoint Test N. |
| **No** | GO to Pinpoint Test T to diagnose the RH steering wheel switches. |

| P3 CHECK THE OPERATION OF THE "PHONE" VOICE COMMAND |
|---|

- Pair a Bluetooth phone with the SYNC® system. For information on the pairing process, refer to the Owner's Literature. If necessary, review the device compatibility list found on syncmyride.com.
- Operate the VOICE steering wheel switch and listen for the SYNC® audible prompt.
- **NOTE:** *Make sure the interior of the vehicle is as quiet as possible. Excessive noise may prevent the system from correctly recognizing spoken commands.*

  Clearly speak the word "phone."

**Does the SYNC® system respond correctly to the "phone" voice command?**

| | |
|---|---|
| **Yes** | For a concern with the voice recognition being inoperative or not operating correctly, the system is operating correctly at this time. The concern may have been caused by an insufficiently quiet vehicle. Delete the just-paired Bluetooth phone from the SYNC® system. For information on deleting a Bluetooth device from the SYNC® system, refer to the Owner's Literature.<br>For a concern with poor quality or no outgoing audio being heard on the outside device during a phone call, GO to P4 |
| **No** | GO to P5. |

| P4 CHECK THE OUTGOING PHONE CALL QUALITY AND VOICE VOLUME |
|---|

- Ensure that the Bluetooth phone paired in step 3 has adequate reception.
- **NOTE:** *Make sure the interior of the vehicle is as quiet as possible. Excessive noise may prevent the call from having clear call quality and acceptable voice volume.*

  Using the SYNC® system, place a phone call. For information on making calls from the SYNC® system, refer to the Owner's Literature.
- Ensure that the called party's phone has adequate reception.
- Instruct the called party to determine if the call quality is clear and of acceptable volume.
- Delete the Bluetooth phone paired in step 3 from the SYNC® system. For information on deleting a Bluetooth device from the SYNC® system, refer to the Owner's Literature.

**Is the outgoing call quality clear and of acceptable volume?**

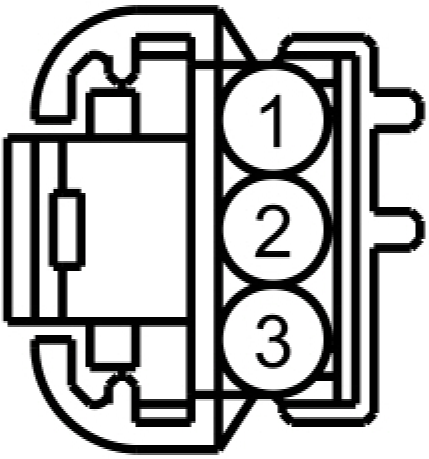
| | |
|---|---|
| **Yes** | The system is operating correctly at this time. The concern may have been caused by poor phone reception or an insufficiently quiet vehicle. |
| **No** | GO to P5. |

| P5 CHECK THE MICROPHONE AND MICROPHONE CIRCUITS FOR CORRECT VOLTAGE DROP |
|---|

**NOTICE: This pinpoint test step directs testing circuits using a back-probe method. Use the special back-probe tool POM6411. Do not force test leads or other probes into connectors. Adequate care must be exercised to avoid connector terminal damage, while ensuring that good electrical contact is made with the circuit or terminal. Failure to follow these instructions may cause damage to wiring, terminals, or connectors, and may cause subsequent electrical faults.**

- Ignition ON.
- While back-probing microphone C9221, measure and record the **DC voltage**.

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C9221 Pin 2 | VMM13 (YE/GN) | C9221 Pin 3 | DMM13 (BK) |

- 


**Is the voltage between 2.7 and 6.2 volts?**

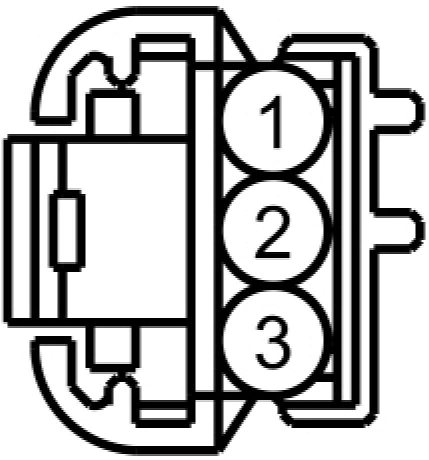
| | |
|---|---|
| **Yes** | If diagnosing an intermittent concern, GO to P6.<br>If **not** diagnosing an intermittent concern, GO to P12. |
| **No** | GO to P7. |

**P6 CHECK THE MICROPHONE CIRCUITS FOR INTERMITTENT CONCERNS**

**NOTICE: This pinpoint test step directs testing circuits using a back-probe method. Use the special back-probe tool POM6411. Do not force test leads or other probes into connectors. Adequate care must be exercised to avoid connector terminal damage, while ensuring that good electrical contact is made with the circuit or terminal. Failure to follow these instructions may cause damage to wiring, terminals, or connectors, and may cause subsequent electrical faults.**

- While back-probing microphone C9221, measure the **DC voltage** while agitating the microphone harness back and forth and up and down (wiggle testing).

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C9221 Pin 2 | VMM13 (YE/GN) | C9221 Pin 3 | DMM13 (BK) |




- Review the recorded results from step 5.

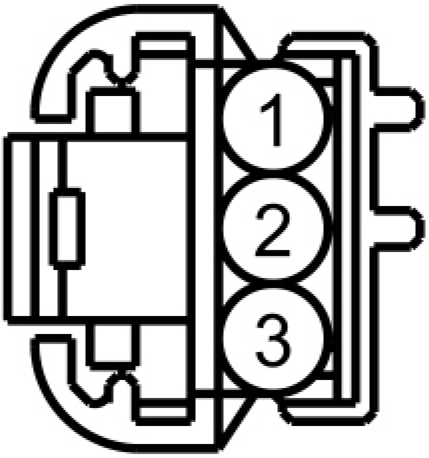
**Does the voltage fluctuate more than .5 volts from the recorded result in step 5?**

| | |
|---|---|
| **Yes** | REPAIR the circuit(s), connector, or terminal(s) in question. For corrosion, clean pins or install a new connector or terminals. For damaged or bent pins, install new terminals or pins. |
| **No** | GO to P11. |

**P7 CHECK THE MICROPHONE CIRCUITS TO THE APIM FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: Microphone C9221.
- Disconnect: APIM C2383.
- Ignition ON.
- Measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C9221 Pin 2 | VMM13 (YE/GN) | — | Ground |
| C9221 Pin 3 | DMM13 (BK) | — | Ground |



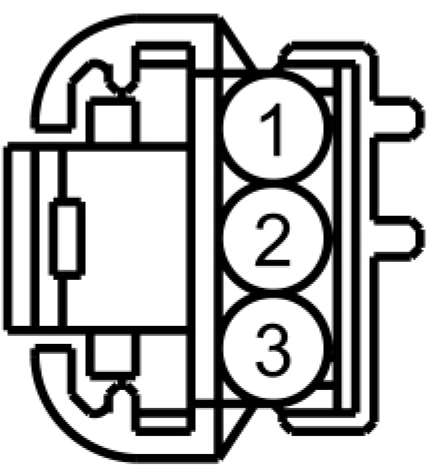

**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit in question. |
| **No** | GO to P8. |

**P8 CHECK THE MICROPHONE CIRCUITS TO THE APIM FOR A SHORT TO GROUND**

- Ignition OFF.
- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C9221 Pin 2 | VMM13 (YE/GN) | — | Ground |
| C9221 Pin 3 | DMM13 (BK) | — | Ground |




**Are the resistances greater than 10,000 ohms?**

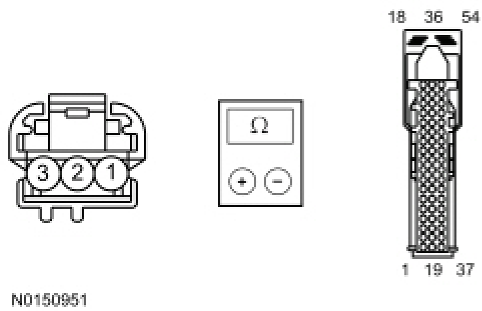
| | |
|---|---|
| **Yes** | GO to P9. |

| | |
|---|---|
| **No** | REPAIR the circuit in question. |

## P9 CHECK THE MICROPHONE CIRCUITS TO THE APIM FOR AN OPEN

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C9221 Pin 2 | VMM13 (YE/GN) | C2383 Pin 5 | VMM13 (YE/GN) |
| C9221 Pin 3 | DMM13 (BK) | C2383 Pin 6 | DMM13 (BK) |

- 

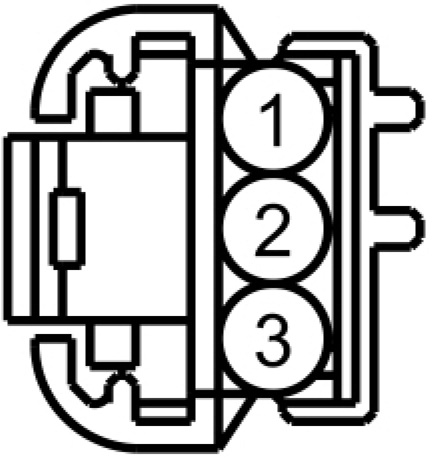

**Are the resistances less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to P10. |
| **No** | REPAIR the circuit in question. |

## P10 CHECK THE MICROPHONE CIRCUITS TO THE APIM FOR A SHORT TOGETHER

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C9221 Pin 2 | VMM13 (YE/GN) | C9221 Pin 3 | DMM13 (BK) |




- 

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to P12. |
| **No** | REPAIR the circuit in question. |

**P11 ISOLATE THE MICROPHONE**

- Install a new microphone. Refer to Microphone.
- If necessary, connect: APIM C2383.
- Operate the audio system in AM/FM mode.
- Operate the VOICE steering wheel switch and listen for the SYNC® audible prompt.
- Speak a command to test the voice recognition for normal operation.

**Is the voice recognition functioning correctly?**

| | |
|---|---|
| **Yes** | The system is operating correctly at this time. The concern was caused by an inoperative microphone. |
| **No** | GO to P12. |

**P12 CHECK FOR CORRECT APIM OPERATION**

- Ignition OFF.
- Disconnect and inspect the APIM connector.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the APIM connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the APIM to re-initialize.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00. |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

### Pinpoint Test Q: The Audio Input Jack Is Inoperative Or Has Poor Sound Quality

#### Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

#### Normal Operation and Fault Conditions

Refer to Audio Input Jack Mode in Information and Entertainment System.

#### Possible Sources

- Customer's device
- Wiring, terminals, or connectors
- Audio input jack
- APIM (with SYNC®)
- ACM (without SYNC®)

**PINPOINT TEST Q : THE AUDIO INPUT JACK IS INOPERATIVE OR HAS POOR SOUND QUALITY**

**NOTE:** *Before carrying out this pinpoint test, make sure the MP3 device is operating correctly.*

**Q1 CHECK THE AUDIO INPUT JACK AUDIO FUNCTIONALITY**

- Connect: Multi-Media Interface Tester 105-00120.
- Using the Multi-Media Interface Tester 105-00120, attempt to play an audio file using the audio input jack.

**Is the audio output OK for the audio input jack?**

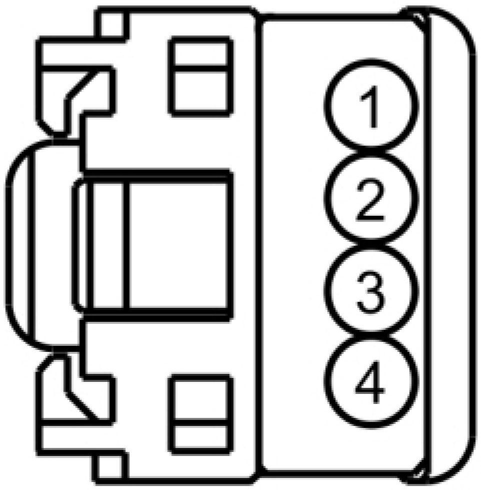
| | |
|---|---|
| **Yes** | The system is operating correctly at this time. The concern may be with the customer's device. |
| **No** | GO to Q2. |

**Q2 CHECK THE AUDIO INPUT JACK CIRCUITS FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: Audio Input Jack C2362.
- Disconnect: APIM C2383 (with SYNC®).

- Disconnect: ACM C290B (without SYNC®).
- Ignition ON.
- Measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C2362 Pin 1 | VME46 (BU/GN) | — | Ground |
| C2362 Pin 3 | RME45 (YE/GN) | — | Ground |
| C2362 Pin 4 | VME45 (BU) | — | Ground |




- 

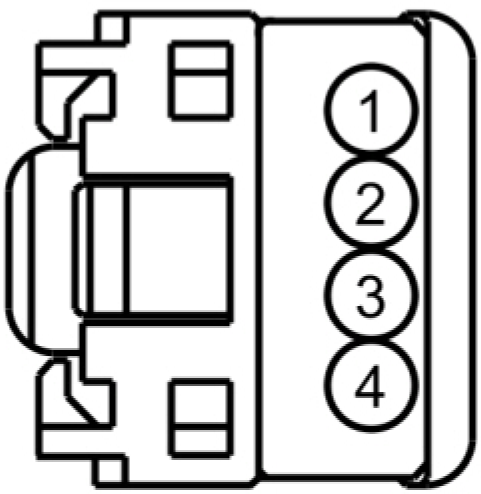
**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit in question. |
| **No** | GO to Q3. |

**Q3 CHECK THE AUDIO INPUT JACK CIRCUITS FOR A SHORT TO GROUND**

- Ignition OFF.
- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C2362 Pin 1 | VME46 (BU/GN) | — | Ground |
| C2362 Pin 3 | RME45 (YE/GN) | — | Ground |
| C2362 Pin 4 | VME45 (BU) | — | Ground |




- 

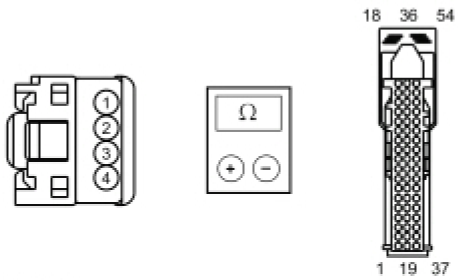
**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to Q4. |
| **No** | REPAIR the circuit. |

**Q4 CHECK THE AUDIO INPUT JACK CIRCUITS FOR AN OPEN**

- For vehicles with SYNC®, measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C2362 Pin 1 | VME46 (BU/GN) | C2383 Pin 47 | VME46 (BU/GN) |
| C2362 Pin 3 | RME45 (YE/GN) | C2383 Pin 46 | RME45 (YE/GN) |
| C2362 Pin 4 | VME45 (BU) | C2383 Pin 45 | VME45 (BU) |

- 

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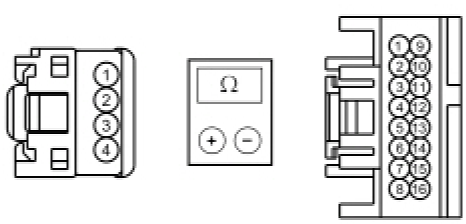
- For vehicles without SYNC®, measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C2362 Pin 1 | VME46 (BU/GN) | C290B Pin 6 | VME46 (BU/GN) |
| C2362 Pin 3 | RME45 (YE/GN) | C290B Pin 8 | RME45 (YE/GN) |
| C2362 Pin 4 | VME45 (BU) | C290B Pin 7 | VME45 (BU) |

- 

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**Are the resistances less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to Q5. |
| **No** | REPAIR the circuit in question. |

**Q5 ISOLATE THE AUDIO INPUT JACK**

- Connect: APIM C2383 (with SYNC®).
- Connect: ACM C290B (without SYNC®).
- Install a new audio input jack. Refer to Audio Input Jack.
- Connect: Multi-Media Interface Tester 105-00120.
- Using the Multi-Media Interface Tester 105-00120, attempt to play an audio file using the audio input jack.

**Is the audio output OK for the audio input jack?**

| | |
|---|---|
| **Yes** | The system is now correctly at this time. The concern was caused by an inoperative audio input jack. |
| **No** | GO to Q6 (with SYNC®).<br>GO to Q7 (without SYNC®). |

**Q6 CHECK FOR CORRECT APIM OPERATION**

- Ignition OFF.
- Disconnect and inspect the APIM connector.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the APIM connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the APIM to re-initialize.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00. |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**Q7 CHECK FOR CORRECT ACM OPERATION**

- Ignition OFF.
- Disconnect and inspect all the ACM connectors.

- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the ACM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test R: The USB Port Is Inoperative Or Does Not Operate Correctly**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

**Normal Operation and Fault Conditions**

Refer to SYNC® AppLink™ , USB Audio Mode, and USB Port. in Information and Entertainment System.

The USB port is part of the USB cable, and cannot be replaced separately.

The Multi-Media Interface Tester 105-00120 is powered by the USB port.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| B1252:04 | USB Port: System Internal Failure | Set by the APIM when an over-temperature condition is detected in the USB circuit. This can be caused by a fault in the USB cable, USB port, or customer's USB device. |
| B1252:11 | USB Port: Circuit Short to Ground | Set by the APIM when a short to ground or over-current condition is detected in the USB circuit. This can be caused by a fault in the USB cable, USB port, or customer's USB device. |

**Possible Sources**

- Customer's device
- USB cable/port
- USB cable/port connection
- APIM

## PINPOINT TEST R : THE USB PORT IS INOPERATIVE OR DOES NOT OPERATE CORRECTLY

### R1 RESET THE APIM AND RECHECK SYNC® SYSTEM OPERATION

- Remove any devices connected to the USB port.
- Carry out an APIM power reset by disconnecting the battery for 1 minute, then reconnecting it.
- Wait at least 2 minutes for the APIM to re-initialize.
- Connect: Multi-Media Interface Tester 105-00120.
- Using the Multi-Media Interface Tester 105-00120, attempt to play an audio file using the USB port.

**Is the audio output OK for the USB port?**

| | |
|---|---|
| **Yes** | The system is operating correctly at this time. The concern was corrected by performing the APIM power reset. |
| **No** | GO to R2. |

### R2 CHECK THE APIM FOR DTC B1252:04 OR DTC B1252:11

- Using a diagnostic scan tool, clear the APIM DTCs.
- Perform the APIM self-test.

**Is DTC B1252:04 or DTC B1252:11 present?**

| | |
|---|---|
| **Yes** | GO to R3. |
| **No** | GO to R5. |

### R3 INSPECT THE USB CABLE AND APIM

- Ignition OFF.
- Disconnect: USB cable at the APIM .
- Inspect the APIM connector port and both ends of the USB cable for damage. Remove any debris.

**Is any damage present?**

| | |
|---|---|
| **Yes** | For the USB cable, INSTALL a new USB cable. REFER to Universal Serial Bus (USB) Cable and Port.<br>For the APIM , INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00. |
| **No** | GO to R4. |

### R4 CHECK THE USB PORT FOR FUNCTIONALITY

- Connect: USB cable at the APIM .
- Connect: Multi-Media Interface Tester 105-00120.

- Using the Multi-Media Interface Tester 105-00120, attempt to play an audio file using the USB port.

**Is the audio output OK for the USB port?**

| | |
|---|---|
| **Yes** | The system is operating correctly at this time. The concern was caused by the USB connector not being seated correctly. |
| **No** | GO to R5. |

| |
|---|
| **R5 ISOLATE THE USB CABLE/PORT** |

- Install a new USB cable/port. Universal Serial Bus (USB) Cable and Port.
- Connect: Multi-Media Interface Tester 105-00120.
- Using the Multi-Media Interface Tester 105-00120, attempt to play a file using the USB port.

**Is the audio output OK for the USB port?**

| | |
|---|---|
| **Yes** | The system is operating correctly at this time. The concern was caused by an inoperative USB cable. |
| **No** | GO to R6. |

| |
|---|
| **R6 CHECK FOR CORRECT APIM OPERATION** |

- Ignition OFF.
- Disconnect and inspect the APIM connector.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the APIM connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the APIM to re-initialize.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00. |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test S: Unable To Pair Bluetooth Device**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

**Normal Operation and Fault Conditions**

Refer to Bluetooth Mode and SYNC® AppLink™ in Information and Entertainment System.

**Possible Sources**

- Incompatible Bluetooth device
- APIM

**PINPOINT TEST S : UNABLE TO PAIR BLUETOOTH DEVICE**

**S1 CHECK THE BLUETOOTH CONNECTION**

**NOTE:** *Carrying out a Master Reset returns all preference settings to the factory defaults, erases the phone book and call histories, and deletes any devices paired with the SYNC® system.*

- Connect: Multi-Media Interface Tester 105-00120.
- Using the Multi-Media Interface Tester 105-00120, connect to the SYNC® system using Bluetooth. Follow the tool instructions.
- Using a diagnostic scan tool, view the APIM PIDs.
- Monitor the Bluetooth device paired PID (BT_PAIR) and the Bluetooth device connected PID (BT_CONN).

**Do the PIDs both read "Yes"?**

| | |
|---|---|
| **Yes** | The SYNC® system is operating correctly at this time. CARRY OUT a SYNC® system Master Reset. REFER to the Owner's Literature.<br>REVIEW the pairing process with the customer. If the customer's device still does not pair, the fault is with the customer's device. |
| **No** | GO to S2. |

**S2 CHECK FOR CORRECT APIM OPERATION**

- Ignition OFF.
- Disconnect and inspect the APIM connector.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the APIM connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the APIM to re-initialize.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00. |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test T: The RH steering wheel controls are inoperative or do not operate correctly**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

**Normal Operation and Fault Conditions**

Refer to Steering Wheel Switch Functions and Steering Wheel Switches in Information and Entertainment System.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| B1380:09 | Steering Wheel Right Switch Pack: Component Failure | Set by the SCCM when the RH steering wheel switch is stuck. |
| B1380:11 | Steering Wheel Right Switch Pack: Circuit Short To Ground | Set by the SCCM when a short to ground is detected in the RH steering wheel switch circuits. |
| B1380:17 | Steering Wheel Right Switch Pack: Circuit Voltage Above Threshold | Set by the SCCM when a short to voltage is detected in the RH steering wheel switch circuits. |

**Possible Sources**

- Steering wheel switch
- Wiring, terminals, or connectors
- Clockspring
- SCCM

**PINPOINT TEST T : THE RH STEERING WHEEL CONTROLS ARE INOPERATIVE OR DO NOT OPERATE CORRECTLY**

**T1 CHECK THE RH STEERING WHEEL SWITCH PIDS**

- Ignition ON.
- Monitor the steering wheel controls PIDs while pressing each RH steering wheel switch.
- Using a diagnostic scan tool, view the SCCM PIDs.
  - SW_VOLDOWN
  - SW_VOLUP

- SW_SEEKMINUS
- SW_SEEKPLUS
- SW_VOICE
- SW_MEDIA
- SW_PHONE

- Using a diagnostic scan tool, view the APIM PID.
  - SW_OK

**Do the PID values agree with the switch positions?**

| | |
|---|---|
| **Yes** | GO to T10. |
| **No** | If one or more, but not all PID values are incorrect or always read one particular switch position, INSTALL a new RH steering wheel switch. REFER to Section 211-05.<br>If all PID values are incorrect or always read one particular switch position, GO to T2. |

**T2 CHECK FOR IPC DTC U0212:00**

- Ignition ON.
- Using a diagnostic scan tool, clear the DTCs from the IPC .
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, perform the IPC self-test.

**Is DTC U0212:00 set in the IPC ?**

| | |
|---|---|
| **Yes** | REFER to Section 413-01. |
| **No** | GO to T3. |

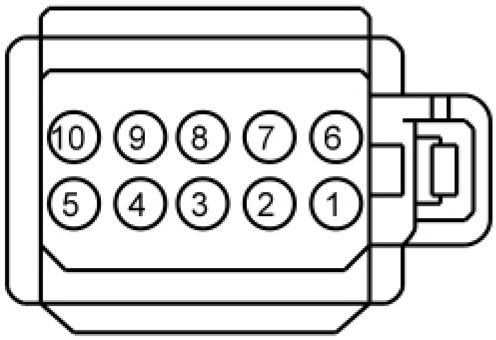
**T3 CHECK THE CIRCUITRY BETWEEN THE CLOCKSPRING AND STEERING WHEEL CONTROLS FOR A SHORT TO GROUND**

**WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Ignition OFF.
- Disconnect: RH Steering Wheel Switch C2999.
- Disconnect: Clockspring C218B.
- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| | | | |

| | | | |
|---|---|---|---|
| C2999 Pin 2 | VME14 | — | Ground |
| C2999 Pin 4 | RME14 | — | Ground |
| C2999 Pin 6 | VME54 | — | Ground |
| C2999 Pin 7 | RME54 | — | Ground |

- 

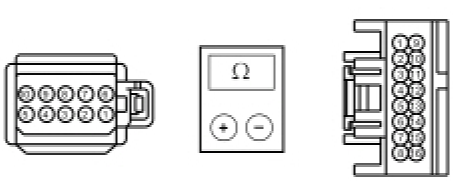

**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to T4. |
| **No** | INSTALL a new steering wheel. REFER to Section 211-04. |

**T4 CHECK THE CIRCUITRY BETWEEN THE CLOCKSPRING AND STEERING WHEEL CONTROLS FOR AN OPEN**

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C2999 Pin 2 | VME14 | C218B Pin 14 | VME14 |
| C2999 Pin 4 | RME14 | C218B Pin 6 | RME14 |
| C2999 Pin 6 | VME54 | C218B Pin 16 | VME54 |
| C2999 Pin 7 | RME54 | C218B Pin 8 | RME54 |

- 

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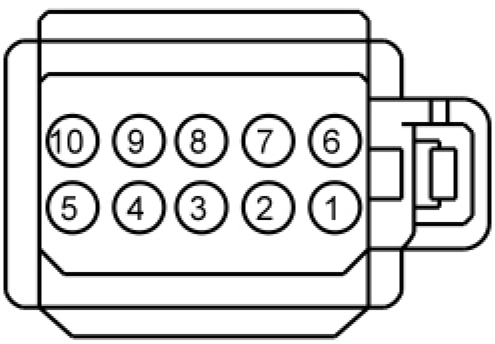
**Are the resistances less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to T5. |
| **No** | INSTALL a new steering wheel. REFER to Section 211-04. |

**T5 CHECK THE CIRCUITRY BETWEEN THE CLOCKSPRING AND STEERING WHEEL CONTROLS FOR A SHORT TOGETHER**

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C2999 Pin 2 | VME14 | C2999 Pin 4 | RME14 |
| C2999 Pin 6 | VME54 | C2999 Pin 7 | RME54 |

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**Are the resistances greater than 10,000 ohms?**

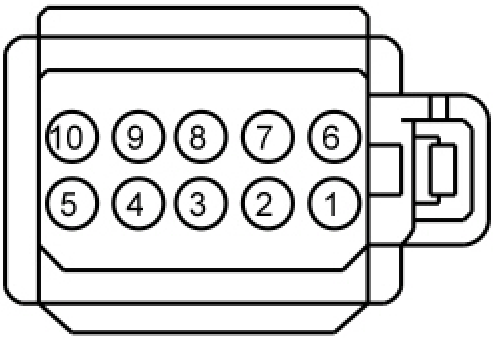
| | |
|---|---|
| **Yes** | GO to T6. |
| **No** | INSTALL a new steering wheel. REFER to Section 211-04. |

**T6 CHECK THE CLOCKSPRING FOR A SHORT TO GROUND**

- Connect: Clockspring C218B.
- Disconnect: Clockspring C218C.
- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |

| | | | |
|---|---|---|---|
| C2999 Pin 2 | VME14 | — | Ground |
| C2999 Pin 4 | RME14 | — | Ground |
| C2999 Pin 6 | VME54 | — | Ground |
| C2999 Pin 7 | RME54 | — | Ground |

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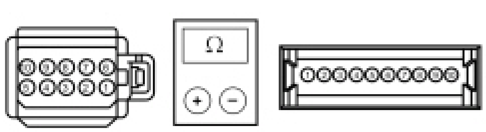
**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to T7. |
| **No** | INSTALL a new clockspring. REFER to Section 501-20B. |

**T7 CHECK THE CLOCKSPRING FOR AN OPEN**

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C2999 Pin 2 | VME14 | C218C Pin 6 | — |
| C2999 Pin 4 | RME14 | C218C Pin 5 | — |
| C2999 Pin 6 | VME54 | C218C Pin 10 | — |
| C2999 Pin 7 | RME54 | C218C Pin 9 | — |

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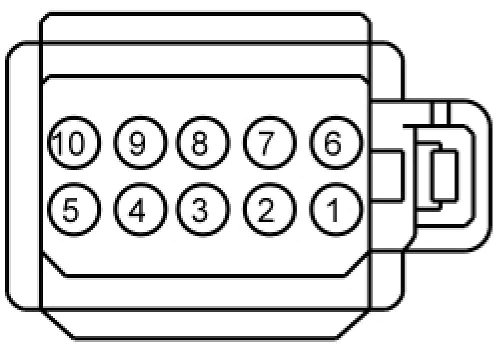
**Are the resistances less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to T8. |
| **No** | INSTALL a new clockspring. REFER to Section 501-20B. |

**T8 CHECK THE CLOCKSPRING FOR A SHORT TOGETHER**

- Measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C2999 Pin 2 | VME14 | C2999 Pin 4 | RME14 |
| C2999 Pin 6 | VME54 | C2999 Pin 7 | RME54 |




-

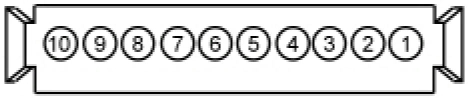
**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to T9. |
| **No** | INSTALL a new clockspring. REFER to Section 501-20B. |

**T9 CHECK FOR VOLTAGE AT THE SCCM**

- Ignition ON.
- Measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C2414D Pin 6 | — | C2414D Pin 5 | — |
| C2414D Pin 10 | — | C2414D Pin 9 | — |




- 

**Is approximately 5 volts present?**

| | |
|---|---|
| **Yes** | INSTALL a new RH steering wheel switch. REFER to Section 211-05. |
| **No** | GO to T10. |

**T10 CHECK FOR CORRECT SCCM OPERATION**

- Ignition OFF.
- Disconnect and inspect all the SCCM connectors.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the SCCM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new SCCM . REFER to Section 211-05. |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test U: The Audio System Does Not Operate Correctly From The FCIM**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

**Normal Operation and Fault Conditions**

Refer to Accessory Delay Feature and FCIM in Information and Entertainment System.

**Possible Sources**

- Network communication concern
- FCIM

**PINPOINT TEST U : THE AUDIO SYSTEM DOES NOT OPERATE CORRECTLY FROM THE FCIM**

**U1 VERIFY THAT THE ACM , APIM , AND FCIM PASS THE NETWORK TEST**

- Using a diagnostic scan tool, perform the network test.

**Did the ACM , APIM , and FCIM pass the network test?**

| | |
|---|---|
| **Yes** | GO to U2. |
| **No** | If the ACM failed the network test, REFER to Section 418-00 The ACM Does Not Respond To The Scan Tool.<br>If the APIM failed the network test, REFER to Section 418-00 The APIM Does Not Respond To The Scan Tool.<br>If the FCIM failed the network test, REFER to Section 418-00 The FCIM Does Not Respond To The Scan Tool. |

**U2 RESET THE AUDIO SYSTEM MODULES AND RECHECK THE FCIM OPERATION**

- Disconnect the battery for 5 minutes, then reconnect it.
- Turn the infotainment system on.
- Test the FCIM for normal operation.

**Does the FCIM operate correctly?**

| | |
|---|---|
| **Yes** | The FCIM is operating correctly at this time. The concern was corrected by resetting the audio system modules. |
| **No** | GO to U3 |

**U3 RETRIEVE THE ACM , APIM , AND FCIM DTCS**

- Using a diagnostic scan tool, perform the self-test for the following modules:
  - ACM
  - APIM
  - FCIM

**Are any DTCs present in the ACM , APIM , or FCIM ?**

| | |
|---|---|
| **Yes** | For ACM DTCs, REFER to the ACM DTC Chart in Information and Entertainment System<br>For APIM DTCs, REFER to the APIM DTC Chart in Information and Entertainment System |

| | |
|---|---|
| | For FCIM DTCs, REFER to the FCIM DTC Chart in Information and Entertainment System |
| **No** | GO to U4. |

| **U4 CHECK FOR CORRECT FCIM OPERATION** |
|---|

- Ignition OFF.
- Disconnect and inspect the FCIM connector.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the FCIM connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test V: The FDIM Is Inoperative Or Does Not Operate Correctly**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

**Normal Operation and Fault Conditions**

Refer to FDIM in Information and Entertainment System.

**Possible Sources**

- Wiring, terminals, or connectors
- FDIM

**PINPOINT TEST V : THE FDIM IS INOPERATIVE OR DOES NOT OPERATE CORRECTLY**

| **V1 VERIFY THAT THE FDIM PASSES THE NETWORK TEST** |
|---|

- Using a diagnostic scan tool, perform the network test.

**Did the FDIM pass the network test?**

| | |
|---|---|
| **Yes** | GO to V2. |
| **No** | REFER to Section 418-00 The FDIM Does Not Respond To The Scan Tool. |

| |
|---|
| **V2 OBSERVE THE FDIM DISPLAY** |

- Turn the audio system on and verify the FDIM operation.

**Is the FDIM screen completely inoperative or blank?**

| | |
|---|---|
| **Yes** | GO to V4. |
| **No** | GO to V3. |

| |
|---|
| **V3 CHECK FOR LOST COMMUNICATION DTCS SET IN THE FDIM** |

- Ignition ON.
- Using a diagnostic scan tool, retrieve all CMDTCs .

**Are any lost communication DTCs (U-codes) present in the FDIM ?**

| | |
|---|---|
| **Yes** | REFER to the FDIM DTC Chart in Information and Entertainment System. |
| **No** | GO to V4. |

| |
|---|
| **V4 CHECK FOR CORRECT FDIM OPERATION** |

- Ignition OFF.
- Disconnect and inspect the FDIM connector.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the FDIM connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new FDIM . REFER to Front Display Interface Module (FDIM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test W: One Or More FDIM Displays Are Inoperative Or Do Not Operate Correctly**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

**Normal Operation and Fault Conditions**

Refer to FDIM in Information and Entertainment System.

**Possible Sources**

- Customer error
- Wiring, terminals, or connectors
- FDIM

**PINPOINT TEST W : ONE OR MORE FDIM DISPLAYS ARE INOPERATIVE OR DO NOT OPERATE CORRECTLY**

**W1 VERIFY THAT THE FDIM PASSES THE NETWORK TEST**

- Using a diagnostic scan tool, perform the network test.

**Did the FDIM pass the network test?**

| | |
|---|---|
| **Yes** | GO to W2. |
| **No** | REFER to Section 418-00 The FDIM Does Not Respond To The Scan Tool. |

**W2 CHECK FOR LOST COMMUNICATION DTCS SET IN THE FDIM**

- Ignition ON.
- Using a diagnostic scan tool, retrieve all CMDTCs .

**Are any lost communication DTCs (U-codes) present in the FDIM ?**

| | |
|---|---|
| **Yes** | REFER to the FDIM DTC Chart in Information and Entertainment System. |
| **No** | GO to W3. |

**W3 CHECK THE FDIM DISPLAY SEGMENTS**

- Perform the Display Test. REFER to Audio Control Module (ACM) Self-Diagnostic Mode

**Do all of the FDIM segments illuminate?**

| | |
|---|---|
| **Yes** | The system is operating correctly at this time. INFORM the customer of the operation of the FDIM display. |
| **No** | GO to W4. |

**W4 CHECK FOR CORRECT FDIM OPERATION**

- Ignition OFF.
- Disconnect and inspect the FDIM connector.

- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the FDIM connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new FDIM . REFER to Front Display Interface Module (FDIM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test X: The Compass Is Inoperative**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

**Normal Operation and Fault Conditions**

Refer to Auto-Dimming Interior Rear View Mirror, Compass Module, and FDIM in Information and Entertainment System.

**DTC Fault Trigger Conditions**

| Root DTC Description | Failure Types | Fault Trigger Conditions |
|---|---|---|
| B1331:02 | Compass/Mirror Module: General Signal Failure | Sets in continuous memory in the BCM when invalid LIN data is received from the compass module. |
| B1331:08 | Compass/Mirror Module: Bus Signal/Message Failure | Sets in continuous memory and on-demand in the BCM when expected LIN messages are not received from the compass module. |
| U0462:82 | Invalid Data Received from Compass Module: Alive/Sequence Counter Incorrect/Not Updated | Set by the IPC when compass data from the BCM is missing or invalid for more than 5 seconds with the ignition in the RUN position. |

If compass data is missing or invalid for 5 seconds or less, the compass display will indicate the vehicle direction based on the last known good compass display message received. When U0462:82 sets, the IPC defaults the compass display to all dashes (---).

**Possible Sources**

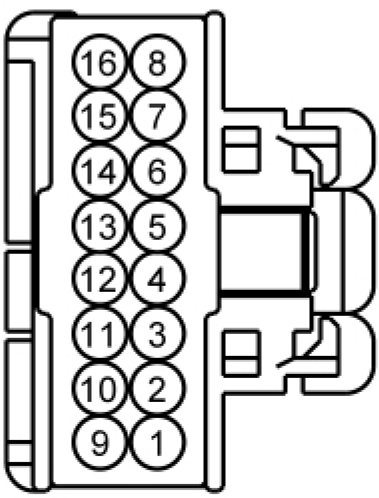
- Wiring, terminals, or connectors
- Compass module
- Auto-dimming interior rear view mirror
- BCM

**PINPOINT TEST X : THE COMPASS IS INOPERATIVE**

**X1 CHECK THE VOLTAGE SUPPLY**

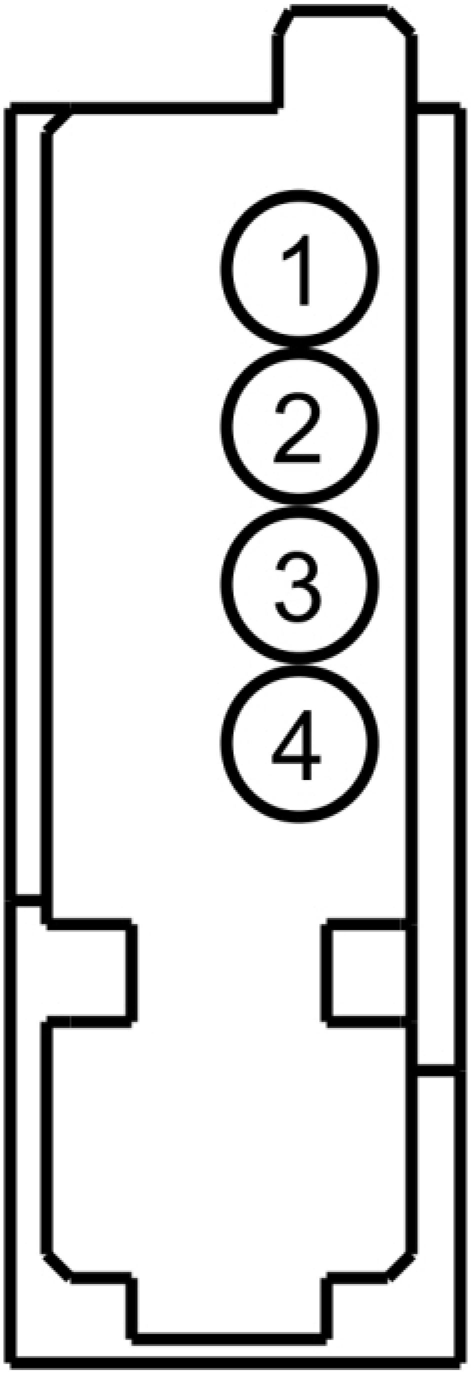
- Ignition OFF.
- Disconnect: Auto-Dimming Interior Rear View Mirror C9039 (With Auto-Dimming Interior Rear View Mirror).
- Disconnect: Compass Module C909 (Without Auto-Dimming Interior Rear View Mirror).
- Ignition ON.
- For vehicles with an auto-dimming interior rear view mirror, measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C9039 Pin 1 | CBP32 (GN/VT) | — | Ground |

- 


- For vehicles **without** an auto-dimming interior rear view mirror, measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C909 Pin 1 | CBP32 (GN/VT) | — | Ground |




- 

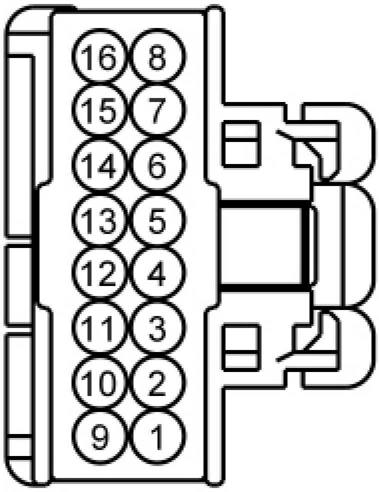
**Is the voltage greater than 11 volts?**

| | |
|---|---|
| **Yes** | GO to X2. |
| **No** | REPAIR the circuit in question. |

**X2 CHECK THE GROUND CIRCUIT**

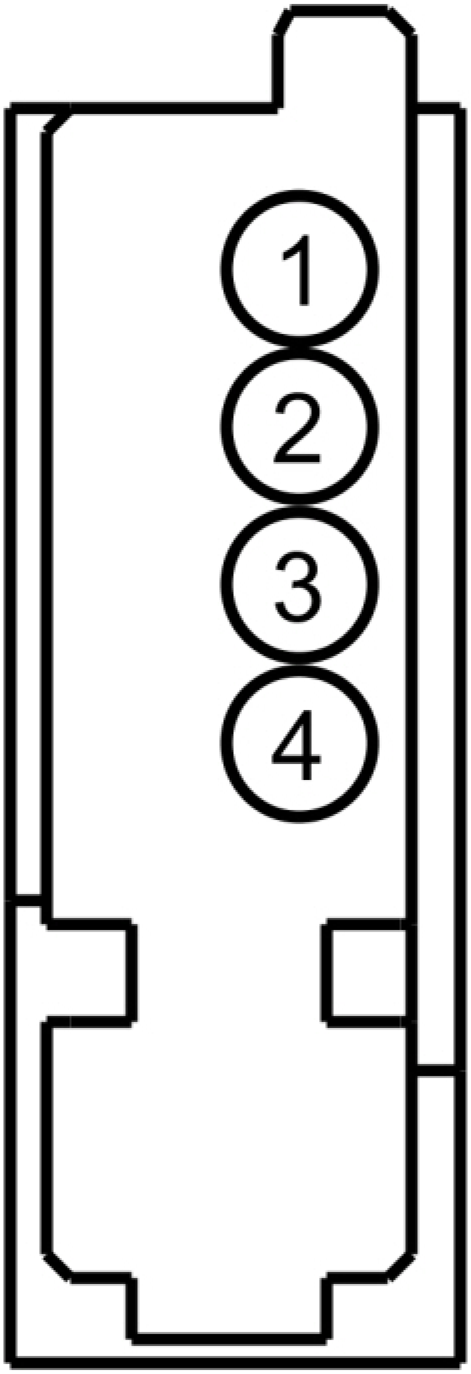
- For vehicles with an auto-dimming interior rear view mirror, measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C9039 Pin 1 | CBP32 (GN/VT) | C9039 Pin 4 | GD133 (BK) |

- 

- For vehicles **without** an auto-dimming interior rear view mirror, measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C909 Pin 1 | CBP32 (GN/VT) | C909 Pin 2 | GD133 (BK) |




- 

**Is the voltage greater than 11 volts?**

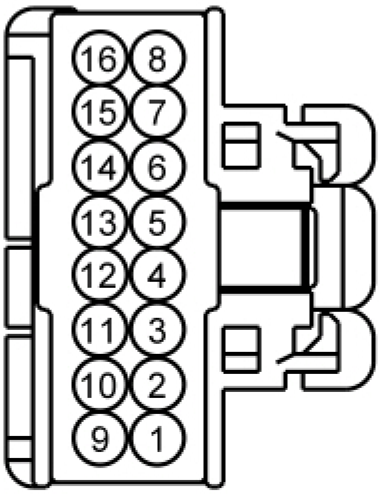
| | |
|---|---|
| **Yes** | GO to X3. |
| **No** | REPAIR the circuit in question. |

**X3 CHECK THE LIN CIRCUIT FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: BCM C2280B.

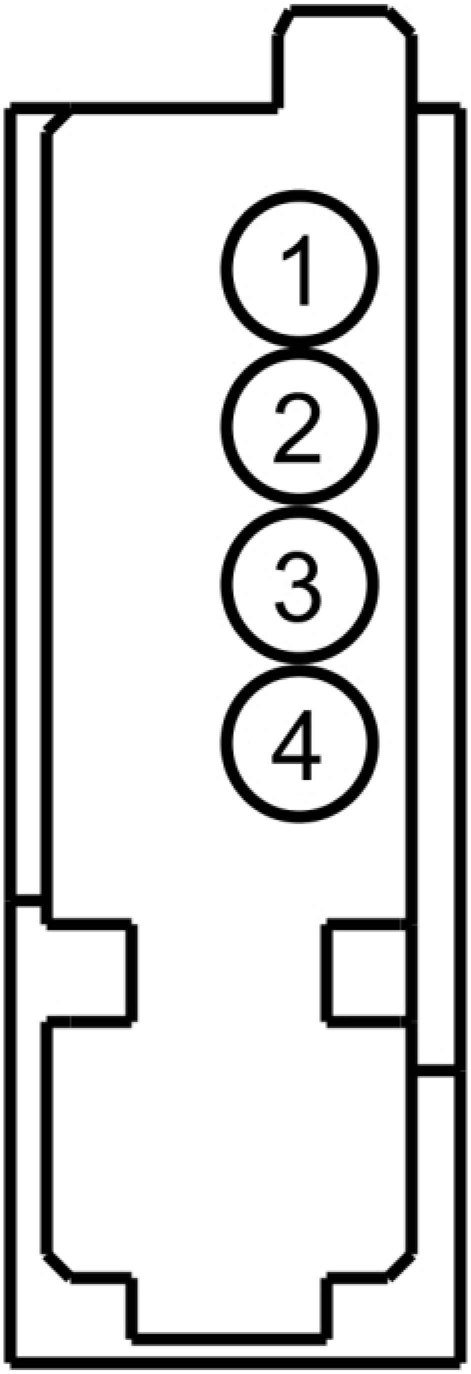
- Ignition ON.
- For vehicles with an auto-dimming interior rear view mirror, measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C9039 Pin 2 | VDN01 (BU/WH) | — | Ground |

- 


- For vehicles **without** an auto-dimming interior rear view mirror, measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C909 Pin 3 | VDN01 (BU/WH) | — | Ground |




- 

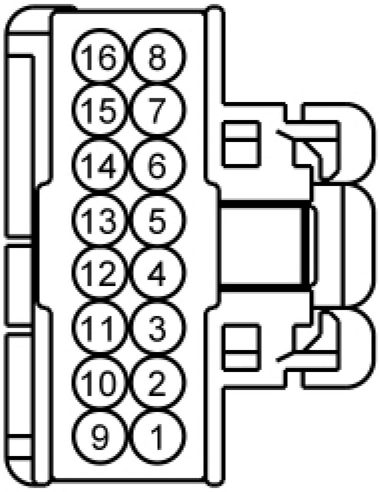
**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit in question. |
| **No** | GO to X4. |

**X4 CHECK THE LIN CIRCUIT FOR A SHORT TO GROUND**

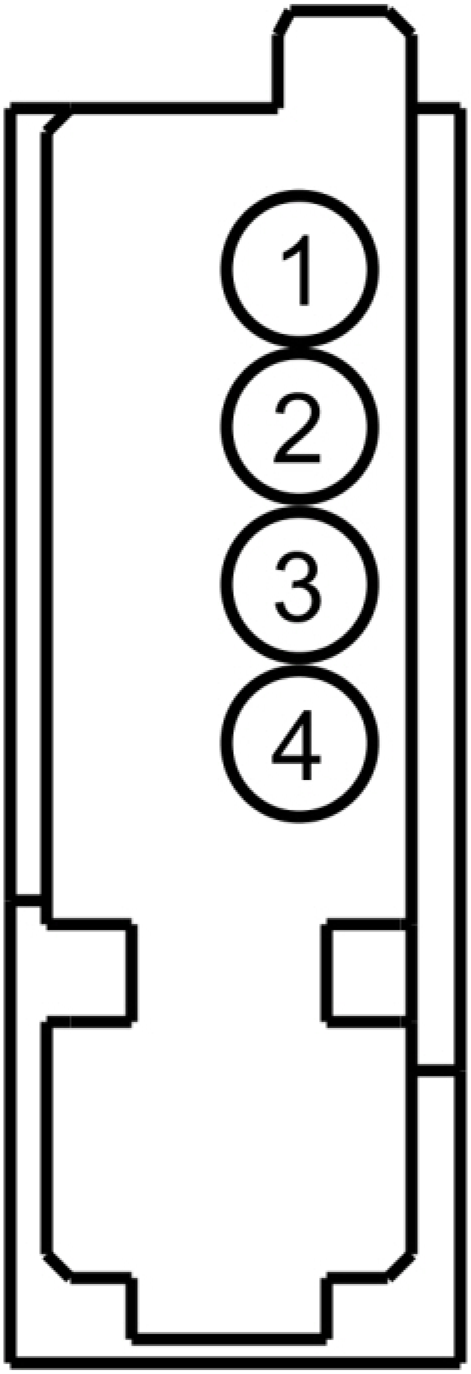
- For vehicles with an auto-dimming interior rear view mirror, measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C9039 Pin 2 | VDN01 (BU/WH) | — | Ground |

- 


- For vehicles **without** an auto-dimming interior rear view mirrorr, measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C909 Pin 3 | VDN01 (BU/WH) | — | Ground |




- 

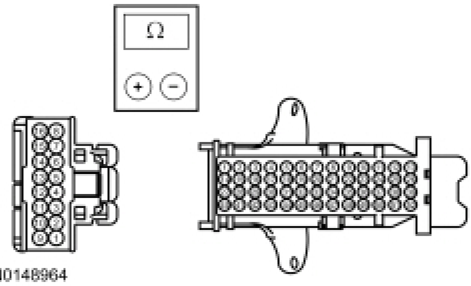
**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to X5. |
| **No** | REPAIR the circuit in question. |

**X5 CHECK THE LIN CIRCUIT FOR AN OPEN**

- For vehicles with an auto-dimming interior rear view mirror, measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C9039 Pin 2 | VDN01 (BU/WH) | C2280B Pin 52 | VDN01 (BU/WH) |

- 

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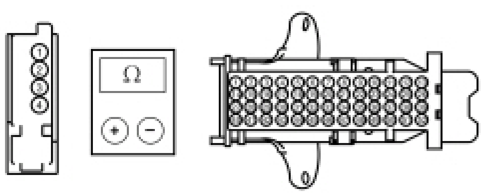
- For vehicles **without** an auto-dimming interior rear view mirror, measure the **resistance** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C909 Pin 3 | VDN01 (BU/WH) | C2280B Pin 52 | VDN01 (BU/WH) |

- 

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**Are the resistances less than 3 ohms?**

| | |
|---|---|
| **Yes** | For vehicles with an auto-dimming interior rear view mirror, GO to X6.<br>For vehicles without an auto-dimming interior rear view mirror, GO to X7. |
| **No** | REPAIR the circuit in question. |

**X6 ISOLATE THE AUTO-DIMMING INTERIOR REAR VIEW MIRROR**

- INSTALL a new auto-dimming interior rear view mirror. REFER to Section 501-09.
- SET the zone. REFER to the Compass Zone Adjustment General Procedure in Section 413-01.
- CALIBRATE the compass. REFER to the Compass Calibration General Procedure in Section 413-01.
- Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | GO to X8. |
| **No** | The system is operating correctly at this time. The concern was corrected by installing a new auto-dimming interior rear view mirror. |

**X7 ISOLATE THE COMPASS MODULE**

- INSTALL a new compass module. REFER to Compass Module.
- SET the zone. REFER to the Compass Zone Adjustment General Procedure in Section 413-01.
- CALIBRATE the compass. REFER to the Compass Calibration General Procedure in Section 413-01.
- Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | GO to X8. |
| **No** | The system is operating correctly at this time. The concern was corrected by installing a new compass module. |

**X8 CHECK FOR CORRECT BCM OPERATION**

- Ignition OFF.
- Disconnect and inspect all the BCM connectors.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the BCM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

### Pinpoint Test Y: The Compass Is Inaccurate

#### Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

**Normal Operation and Fault Conditions**

Refer to Auto-Dimming Interior Rear View Mirror, Compass Module, and FDIM in Information and Entertainment System.

**Possible Sources**

- Zone setting
- Calibration
- Auto-dimming interior mirror
- Compass module

**PINPOINT TEST Y : THE COMPASS IS INACCURATE**

**Y1 CHECK THE COMPASS ACCURACY**

- Position the vehicle and observe the compass display as follows:

| Direction | Compass Display |
|---|---|
| North | N |
| Northeast | NE |
| East | E |
| Southeast | SE |
| South | S |
| Southwest | SW |
| West | W |
| Northwest | NW |

**Does the compass heading accurately display the vehicle direction?**

| | |
|---|---|
| **Yes** | The compass is operating correctly at this time. |
| **No** | GO to Y2. |

**Y2 CHECK THE COMPASS ZONE SETTING AND CALIBRATION**

- SET the zone. REFER to the Compass Zone Adjustment General Procedure in Section 413-01.
- CALIBRATE the compass. REFER to the Compass Calibration General Procedure in Section 413-01.
- Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | The system is operating correctly at this time. The concern may have been caused by an incorrect compass zone selection or compass calibration. |
| **No** | For vehicles with an auto-dimming interior rear view mirror, GO to Y3.<br>For vehicles without an auto-dimming interior rear view mirror, GO to Y4. |

### Y3 ISOLATE THE AUTO-DIMMING INTERIOR REAR VIEW MIRROR

- INSTALL a new auto-dimming interior rear view mirror. REFER to Section 501-09.
- SET the zone. REFER to the Compass Zone Adjustment General Procedure in Section 413-01.
- CALIBRATE the compass. REFER to the Compass Calibration General Procedure in Section 413-01.
- Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | GO to Y5. |
| **No** | The system is operating correctly at this time. The concern was corrected by installing a new auto-dimming interior rear view mirror. |

### Y4 ISOLATE THE COMPASS MODULE

- INSTALL a new compass module. REFER to Compass Module.
- SET the zone. REFER to the Compass Zone Adjustment General Procedure in Section 413-01.
- CALIBRATE the compass. REFER to the Compass Calibration General Procedure in Section 413-01.
- Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | GO to Y5. |
| **No** | The system is operating correctly at this time. The concern was corrected by installing a new compass module. |

### Y5 CHECK FOR CORRECT BCM OPERATION

- Ignition OFF.
- Disconnect all the BCM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. |

**Pinpoint Test Z: DTC B1318**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System for schematic and connector information.

**Normal Operation and Fault Conditions**

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| B1318 | Battery Voltage Low | Set by the FDIM when the supply voltage falls below 10 volts for at least 10 seconds during normal operation or for more than 250 milliseconds during the self-test. |

**PINPOINT TEST Z : DTC B1318**

**Z1 RECHECK FOR DTC B1318**

- Ignition ON.
- Using a diagnostic scan tool, clear the FDIM DTCs.
- Wait at least 15 seconds.
- Using a diagnostic scan tool, repeat the self-test.

**Is DTC B1318 still present?**

| | |
|---|---|
| **Yes** | GO to Z2. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to a previous low battery voltage condition. |

**Z2 CHECK FOR CHARGING SYSTEM DTCS IN THE PCM**

- Using a diagnostic scan tool, retrieve all CMDTCs .

**Are any voltage-related DTCs set in the PCM ?**

| | |
|---|---|
| **Yes** | REFER to PCM DTC Chart in Section 303-14. |
| **No** | GO to Z3. |

**Z3 CHECK THE BATTERY CONDITION AND STATE OF CHARGE**

- Check the battery condition and verify that the battery is fully charged. REFER to Battery Condition Test in Section 414-01.

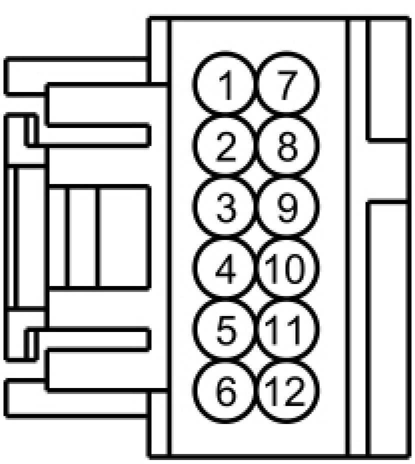
**Is the battery OK and fully charged?**

| | |
|---|---|
| **Yes** | GO to Z4. |
| **No** | REFER to the Symptom Chart in Section 414-01. |

**Z4 CHECK THE FDIM VOLTAGE SUPPLY**

- Ignition OFF.
- Measure and record the voltage at the battery.
- Disconnect: FDIM C2123.
- Ignition ON.
- Measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C2123 Pin 12 | SBP09 (RD) | — | Ground |

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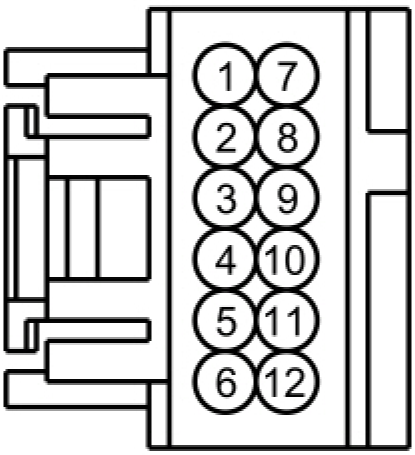
**Is the voltage within 0.2 volts of the recorded battery voltage?**

| | |
|---|---|
| **Yes** | GO to Z5. |
| **No** | REPAIR the circuit for high resistance. |

**Z5 CHECK THE FDIM GROUND CIRCUIT FOR CONTINUITY**

- Measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C2123 Pin 12 | SBP09 (RD) | C2123 Pin 8 | GD133 (BK) |

- 


**Is the voltage greater than 11 volts?**

| | |
|---|---|
| **Yes** | GO to Z6. |
| **No** | REPAIR the circuit for high resistance. |

**Z6 CHECK FOR CORRECT FDIM OPERATION**

- Ignition OFF.
- Disconnect and inspect the FDIM connector.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the FDIM connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new FDIM . REFER to Front Display Interface Module (FDIM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test AA: DTC U0100:00**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| U0100:00 | Lost Communication with ECM/PCM "A": No Sub Type Information | Set by the GPSM when network messages are missing from the PCM for more than 5 seconds with the ignition in RUN. |

**PINPOINT TEST AA : DTC U0100:00**

**AA1 VERIFY THE CUSTOMER'S CONCERN**

- Ignition ON.
- Verify if there is an observable symptom present.

**Is an observable symptom present?**

| | |
|---|---|
| **Yes** | GO to AA2. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AA2 RECHECK FOR DTC U0100:00**

- Ignition ON.
- Using a diagnostic scan tool, clear all CMDTCs .
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, retrieve all CMDTCs .

**Is DTC U0100:00 set in the GPSM ?**

| | |
|---|---|
| **Yes** | GO to AA3. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AA3 CHECK THE PCM FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS**

- Using a diagnostic scan tool, perform the PCM self-test.

**Are any low or high voltage DTCs recorded in the PCM ?**

| | |
|---|---|
| **Yes** | REFER to PCM DTC Chart in Section 303-14. |
| **No** | GO to AA4. |

**AA4 CHECK THE GPSM FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS**

- Using a diagnostic scan tool, perform the GPSM self-test.

**Are any low or high voltage DTCs recorded in the GPSM ?**

| | |
|---|---|
| **Yes** | For DTC U3003:16, GO to Pinpoint Test AN<br>For DTC U3003:17, GO to Pinpoint Test AO |
| **No** | GO to AA5. |

**AA5 CHECK FOR CORRECT PCM OPERATION**

- Ignition OFF.
- Disconnect and inspect all the PCM connectors.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the PCM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new PCM . REFER to Section 303-14. |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

## Pinpoint Test AB: DTC U0140

### Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

### DTC Fault Trigger Conditions

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| U0140 | Lost Communication with Body Control Module (GEM) | Set by the FDIM when network messages are missing from the BCM for more than 5 seconds with the ignition in RUN. |

**PINPOINT TEST AB : DTC U0140**

**AB1 VERIFY THE CUSTOMER'S CONCERN**

- Ignition ON.
- Verify if there is an observable symptom present.

**Is an observable symptom present?**

| | |
|---|---|
| **Yes** | GO to AB2. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AB2 RECHECK FOR DTC U0140**

- Ignition ON.
- Using a diagnostic scan tool, clear all CMDTCs .
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, retrieve all CMDTCs .

**Is DTC U0140 set in the FDIM ?**

| | |
|---|---|
| **Yes** | GO to AB3. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

| AB3 CHECK THE BCM FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS |
|---|

- Using a diagnostic scan tool, perform the BCM self-test.

**Are any low or high voltage DTCs recorded in the BCM ?**

| | |
|---|---|
| **Yes** | REFER to BCM DTC Chart in Section 419-10. |
| **No** | GO to AB4. |

| AB4 CHECK FOR CORRECT BCM OPERATION |
|---|

- Ignition OFF.
- Disconnect and inspect all the BCM connectors.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the BCM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test AC: DTC U0140:00**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| U0140:00 | Lost Communication With Body Control Module: No Sub Type Information | Set by the ACM , APIM (if equipped), FCIM , and GPSM when network messages are missing from the BCM for more than 5 seconds with the ignition in RUN. |

**PINPOINT TEST AC : DTC U0140:00**

**AC1 VERIFY THE CUSTOMER'S CONCERN**

- Ignition ON.
- Verify if there is an observable symptom present.

**Is an observable symptom present?**

| | |
|---|---|
| **Yes** | GO to AC2. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AC2 RECHECK FOR DTC U0140:00**

- Ignition ON.
- Using a diagnostic scan tool, clear all CMDTCs .
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, retrieve all CMDTCs .

**Is DTC U0140:00 set in the ACM , APIM (if equipped), FCIM , or GPSM ?**

| | |
|---|---|
| **Yes** | GO to AC3. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AC3 CHECK THE BCM FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS**

- Using a diagnostic scan tool, perform the BCM self-test.

**Are any low or high voltage DTCs recorded in the BCM ?**

| | |
|---|---|
| **Yes** | REFER to BCM DTC Chart in Section 419-10. |
| **No** | GO to AC4. |

**AC4 CHECK FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS**

- Using a diagnostic scan tool, perform the self-test for the following modules:
  - ACM
  - APIM (if equipped)
  - FCIM
  - GPSM

**Are low or high voltage DTCs recorded in the ACM , APIM (if equipped), FCIM , or GPSM ?**

| | |
|---|---|
| **Yes** | For DTC U3003:16, GO to Pinpoint Test AN<br>For DTC U3003:17, GO to Pinpoint Test AO |
| **No** | GO to AC5. |

**AC5 CHECK FOR DTC U0140:00 SET IN MULTIPLE MODULES**

- Review the recorded results from the self-test.

**Is DTC U0140:00 set in more than one audio system module?**

| | |
|---|---|
| **Yes** | GO to AC6. |
| **No** | If there is an observable symptom, GO to Symptom Chart — General Audio System to diagnose the observed symptom.<br>If there is no observable symptom, the system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AC6 CHECK FOR CORRECT BCM OPERATION**

- Ignition OFF.
- Disconnect and inspect all the BCM connectors.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the BCM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test AD: DTC U0155**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| U0155 | Lost Communication with Instrument Panel Cluster (IPC ) Control Panel | Set by the FDIM when network messages are missing from the IPC for more than 5 seconds with the ignition in RUN. |

**PINPOINT TEST AD : DTC U0155**

**AD1 VERIFY THE CUSTOMER'S CONCERN**

- Ignition ON.
- Verify if there is an observable symptom present.

**Is an observable symptom present?**

| | |
|---|---|
| **Yes** | GO to AD2. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AD2 RECHECK FOR DTC U0155**

- Ignition ON.
- Using a diagnostic scan tool, clear all CMDTCs .
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, retrieve all CMDTCs .

**Is DTC U0155 set in the FDIM ?**

| | |
|---|---|
| **Yes** | GO to AD3. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AD3 CHECK THE IPC FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS**

- Using a diagnostic scan tool, perform the IPC self-test.

**Are any low or high voltage DTCs recorded in the IPC ?**

| | |
|---|---|
| **Yes** | REFER to IPC DTC Chart in Section 413-01. |
| **No** | GO to AD4. |

| AD4 CHECK FOR CORRECT IPC OPERATION |
|---|
| • Ignition OFF.<br>• Disconnect and inspect the IPC connector.<br>• Repair:<br>  ▪ Corrosion (clean module pins or install new connectors or terminals)<br>  ▪ Damaged or bent pins (install new terminals or pins)<br>  ▪ Pushed-out pins (install new pins as necessary)<br>• Reconnect the IPC connector. Make sure it seats and latches correctly.<br>• Operate the system and determine if the concern is still present. |

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new IPC . REFER to Section 413-01. |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

## Pinpoint Test AE: DTC U0155:00

### Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

### DTC Fault Trigger Conditions

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| U0155:00 | Lost Communication With Instrument Panel Cluster (IPC ) Control Module: No Sub Type Information | Set by the ACM , APIM (if equipped), and GPSM when network messages are missing from the IPC for more than 5 seconds with the ignition in RUN. |

**PINPOINT TEST AE : DTC U0155:00**

| AE1 VERIFY THE CUSTOMER'S CONCERN |
|---|
| • Ignition ON. |

- Verify if there is an observable symptom present.

**Is an observable symptom present?**

| | |
|---|---|
| **Yes** | GO to AE2. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AE2 RECHECK FOR DTC U0155:00**

- Ignition ON.
- Using a diagnostic scan tool, clear all CMDTCs .
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, retrieve all CMDTCs .

**Is DTC U0155:00 set in the ACM , APIM (if equipped), or GPSM ?**

| | |
|---|---|
| **Yes** | GO to AE3. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AE3 CHECK THE IPC FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS**

- Using a diagnostic scan tool, perform the IPC self-test.

**Are any low or high voltage DTCs recorded in the IPC ?**

| | |
|---|---|
| **Yes** | REFER to IPC DTC Chart in Section 413-01. |
| **No** | GO to AE4. |

**AE4 CHECK FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS**

- Using a diagnostic scan tool, perform the self-test for the following modules:
  - ACM
  - APIM (if equipped)
  - GPSM

**Are low or high voltage DTCs recorded in the ACM , APIM (if equipped), or GPSM ?**

| | |
|---|---|
| **Yes** | For DTC U3003:16, GO to Pinpoint Test AN<br>For DTC U3003:17, GO to Pinpoint Test AO |

| | |
|---|---|
| **No** | GO to AE5. |

**AE5 CHECK FOR DTC U0155:00 SET IN MULTIPLE MODULES**

- Review the recorded results from the self-test.

**Is DTC U0155:00 set in more than one audio system module?**

| | |
|---|---|
| **Yes** | GO to AE6. |
| **No** | If there is an observable symptom, GO to Symptom Chart — General Audio System to diagnose the observed symptom.<br>If there is no observable symptom, the system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AE6 CHECK FOR CORRECT IPC OPERATION**

- Ignition OFF.
- Disconnect and inspect the IPC connector.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the IPC connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new IPC . REFER to Section 413-01. |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

## Pinpoint Test AF: DTC U0159:00

### Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

### DTC Fault Trigger Conditions

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| U0159:00 | Lost Communication With Parking Assist Control Module "A": No Sub Type Information | Set by the ACM when the volume cutback message is missing from the PAM for more than 5 seconds with the ignition in RUN. When the message is missing, the ACM defaults to no volume cutback when the parking aid tone is sounding. |

## PINPOINT TEST AF : DTC U0159:00

### AF1 VERIFY THE CUSTOMER'S CONCERN

- Ignition ON.
- Verify if there is an observable symptom present.

**Is an observable symptom present?**

| | |
|---|---|
| **Yes** | GO to AF2. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

### AF2 RECHECK FOR DTC U0159:00

- Ignition ON.
- Using a diagnostic scan tool, clear all CMDTCs .
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, retrieve all CMDTCs .

**Is DTC U0159:00 set in the ACM ?**

| | |
|---|---|
| **Yes** | GO to AF3. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

### AF3 CHECK THE PAM FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the PAM self-test.

**Are any low or high voltage DTCs recorded in the PAM ?**

| | |
|---|---|
| **Yes** | REFER to PAM DTC Chart in Section 413-13. |
| **No** | GO to AF4. |

| AF4 CHECK THE ACM FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS |
|---|
| • Using a diagnostic scan tool, perform the ACM self-test.<br><br>**Are any low or high voltage DTCs recorded in the ACM ?** |

| | |
|---|---|
| **Yes** | For DTC U3003:16, GO to Pinpoint Test AN<br>For DTC U3003:17, GO to Pinpoint Test AO |
| **No** | GO to AF5. |

| AF5 CHECK FOR CORRECT PAM OPERATION |
|---|
| • Ignition OFF.<br>• Disconnect and inspect the PAM connector.<br>• Repair:<br>▪ Corrosion (clean module pins or install new connectors or terminals)<br>▪ Damaged or bent pins (install new terminals or pins)<br>▪ Pushed-out pins (install new pins as necessary)<br>• Reconnect the PAM connector. Make sure it seats and latches correctly.<br>• Operate the system and determine if the concern is still present.<br><br>**Is the concern still present?** |

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new PAM . REFER to Section 413-13. |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test AG: DTC U0164**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| U0164 | Lost Communication With HVAC Control Module (EATC) | Set by the FDIM when network messages are missing from the HVAC module for more than 5 seconds with the ignition in RUN. . |

## PINPOINT TEST AG : DTC U0164

### AG1 VERIFY THE CUSTOMER'S CONCERN

- Ignition ON.
- Verify if there is an observable symptom present.

**Is an observable symptom present?**

| | |
|---|---|
| **Yes** | GO to AG2. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

### AG2 RECHECK FOR DTC U0164

- Ignition ON.
- Using a diagnostic scan tool, clear all CMDTCs .
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, retrieve all CMDTCs .

**Is DTC U0164 set in the FDIM ?**

| | |
|---|---|
| **Yes** | GO to AG3. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

### AG3 CHECK THE HVAC MODULE FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the HVAC module self-test.

**Are any low or high voltage DTCs recorded in the HVAC module?**

| | |
|---|---|
| **Yes** | REFER to HVAC module DTC Chart in Section 412-00. |
| **No** | GO to AG4. |

### AG4 CHECK FOR CORRECT HVAC MODULE OPERATION

- Ignition OFF.
- Disconnect and inspect all the HVAC module connectors.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)

- Pushed-out pins (install new pins as necessary)

- Reconnect the HVAC module connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new HVAC module. REFER to Section 412-01. |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test AH: DTC U0184**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| U0184 | Lost Communication With Radio (ACM ) | Set by the FDIM when network messages are missing from the ACM for more than 5 seconds with the ignition in RUN. |

**PINPOINT TEST AH : DTC U0184**

**AH1 VERIFY THE CUSTOMER'S CONCERN**

- Ignition ON.
- Verify if there is an observable symptom present.

**Is an observable symptom present?**

| | |
|---|---|
| **Yes** | GO to AH2. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AH2 RECHECK FOR DTC U0184**

- Ignition ON.
- Using a diagnostic scan tool, clear all CMDTCs .

- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, retrieve all CMDTCs .

**Is DTC U0184 set in the FDIM ?**

| | |
|---|---|
| **Yes** | GO to AH3. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AH3 CHECK THE ACM FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS**

- Using a diagnostic scan tool, perform the ACM self-test.

**Are any low or high voltage DTCs recorded in the ACM ?**

| | |
|---|---|
| **Yes** | For DTC U3003:16, GO to Pinpoint Test AN<br>For DTC U3003:17, GO to Pinpoint Test AO |
| **No** | GO to AH4. |

**AH4 CHECK FOR CORRECT ACM OPERATION**

- Ignition OFF.
- Disconnect and inspect all the ACM connectors.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the ACM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test AI: DTC U0184:00**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| U0184:00 | Lost Communication With Radio: No Sub Type Information | Set by the FCIM when network messages are missing from the ACM for more than 5 seconds with the ignition in RUN. |

**PINPOINT TEST AI : DTC U0184:00**

**AI1 VERIFY THE CUSTOMER'S CONCERN**

- Ignition ON.
- Verify if there is an observable symptom present.

**Is an observable symptom present?**

| | |
|---|---|
| **Yes** | GO to AI2. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AI2 RECHECK FOR DTC U0184:00**

- Ignition ON.
- Using a diagnostic scan tool, clear all CMDTCs .
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, retrieve all CMDTCs .

**Is DTC U0184:00 set in the FCIM ?**

| | |
|---|---|
| **Yes** | GO to AI3. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AI3 CHECK FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS**

- Using a diagnostic scan tool, perform the self-test for the following modules:
  - ACM

- FCIM

**Are low or high voltage DTCs recorded in the ACM or FCIM ?**

| | |
|---|---|
| **Yes** | For DTC U3003:16, GO to Pinpoint Test AN<br>For DTC U3003:17, GO to Pinpoint Test AO |
| **No** | GO to AI4. |

**AI4 CHECK FOR CORRECT ACM OPERATION**

- Ignition OFF.
- Disconnect and inspect all the ACM connectors.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the ACM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

## Pinpoint Test AJ: DTC U0197

### Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

### DTC Fault Trigger Conditions

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| U0197 | Lost Communication with Telephone Control Module | Set by the FDIM when network messages are missing from the APIM (if equipped) for more than 5 seconds with the ignition in RUN. |

**PINPOINT TEST AJ : DTC U0197**

## AJ1 VERIFY THE CUSTOMER'S CONCERN

- Ignition ON.
- Verify if there is an observable symptom present.

**Is an observable symptom present?**

| | |
|---|---|
| **Yes** | GO to AJ2. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

## AJ2 RECHECK FOR DTC U0197

- Ignition ON.
- Using a diagnostic scan tool, clear all CMDTCs .
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, retrieve all CMDTCs .

**Is DTC U0197 set in the FDIM ?**

| | |
|---|---|
| **Yes** | GO to AJ3. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

## AJ3 CHECK THE APIM FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the APIM self-test.

**Are any low or high voltage DTCs recorded in the APIM ?**

| | |
|---|---|
| **Yes** | For DTC U3003:16, GO to Pinpoint Test AN<br>For DTC U3003:17, GO to Pinpoint Test AO |
| **No** | GO to AJ4. |

## AJ4 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the APIM connector.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)

- Pushed-out pins (install new pins as necessary)

- Reconnect the APIM connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the APIM to re-initialize.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00. |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

### Pinpoint Test AK: DTC U0197:00

### Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

### DTC Fault Trigger Conditions

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| U0197:00 | Lost Communication With Telephone Control Module: No Sub Type Information | Set by the ACM when network messages are missing from the APIM (if equipped) for more than 5 seconds with the ignition in RUN. |

**PINPOINT TEST AK : DTC U0197:00**

**AK1 VERIFY THE CUSTOMER'S CONCERN**

- Ignition ON.
- Verify if there is an observable symptom present.

**Is an observable symptom present?**

| | |
|---|---|
| **Yes** | GO to AK2. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AK2 RECHECK FOR DTC U0197:00**

- Ignition ON.

- Using a diagnostic scan tool, clear all CMDTCs .
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, retrieve all CMDTCs .

**Is DTC U0197:00 set in the ACM ?**

| | |
|---|---|
| **Yes** | GO to AK3. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AK3 CHECK FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS**

- Using a diagnostic scan tool, perform the self-test for the following modules:
  - ACM
  - APIM

**Are low or high voltage DTCs recorded in the ACM or APIM ?**

| | |
|---|---|
| **Yes** | For DTC U3003:16, GO to Pinpoint Test AN<br>For DTC U3003:17, GO to Pinpoint Test AO |
| **No** | GO to AK4. |

**AK4 CHECK FOR CORRECT APIM OPERATION**

- Ignition OFF.
- Disconnect and inspect the APIM connector.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the APIM connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the APIM to re-initialize.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00. |
| | |

| | |
|---|---|
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test AL: DTC U0255:00**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| U0255:00 | Lost Communication With Front Display Interface Module: No Sub Type Information | Set by the ACM when network messages are missing from the FDIM for greater than 5 seconds with the ignition in RUN. |

**PINPOINT TEST AL : DTC U0255:00**

**AL1 VERIFY THE CUSTOMER'S CONCERN**

- Ignition ON.
- Verify if there is an observable symptom present.

**Is an observable symptom present?**

| | |
|---|---|
| **Yes** | GO to AL2. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AL2 RECHECK FOR DTC U0255:00**

- Ignition ON.
- Using a diagnostic scan tool, clear all CMDTCs .
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, retrieve all CMDTCs .

**Is DTC U0255:00 set in the ACM ?**

| | |
|---|---|
| **Yes** | GO to AL3. |

| | |
|---|---|
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AL3 CHECK THE ACM FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS**

- Using a diagnostic scan tool, perform the ACM self-test.

**Are any low or high voltage DTCs recorded in the ACM ?**

| | |
|---|---|
| **Yes** | For DTC U3003:16, GO to Pinpoint Test AN<br>For DTC U3003:17, GO to Pinpoint Test AO |
| **No** | GO to AL4. |

**AL4 CHECK FOR CORRECT FDIM OPERATION**

- Ignition OFF.
- Disconnect and inspect the FDIM connector.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the FDIM connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new FDIM . REFER to Front Display Interface Module (FDIM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

## Pinpoint Test AM: DTC U0256:00

### Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

### DTC Fault Trigger Conditions

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| U0256:00 | Lost Communication With Front Controls Interface Module "A": No Sub Type Information | Set by the ACM when network messages are missing from the FCIM for more than 5 seconds with the ignition in the RUN position. |

**PINPOINT TEST AM : DTC U0256:00**

**AM1 VERIFY THE CUSTOMER'S CONCERN**

- Ignition ON.
- Verify if there is an observable symptom present.

**Is an observable symptom present?**

| | |
|---|---|
| **Yes** | GO to AM2. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AM2 RECHECK FOR DTC U0256:00**

- Ignition ON.
- Using a diagnostic scan tool, clear all CMDTCs .
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, retrieve all CMDTCs .

**Is DTC U0256:00 set in the ACM ?**

| | |
|---|---|
| **Yes** | GO to AM3. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AM3 CHECK FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS**

- Using a diagnostic scan tool, perform the self-test for the following modules:
  - ACM
  - FCIM

**Are low or high voltage DTCs recorded in the ACM or FCIM ?**

| | |
|---|---|
| **Yes** | For DTC U3003:16, GO to Pinpoint Test AN<br>For DTC U3003:17, GO to Pinpoint Test AO |
| **No** | GO to AM4. |

| AM4 CHECK FOR CORRECT FCIM OPERATION |
|---|
| <ul><li>Ignition OFF.</li><li>Disconnect and inspect the FCIM connector.</li><li>Repair:<ul><li>Corrosion (clean module pins or install new connectors or terminals)</li><li>Damaged or bent pins (install new terminals or pins)</li><li>Pushed-out pins (install new pins as necessary)</li></ul></li><li>Reconnect the FCIM connector. Make sure it seats and latches correctly.</li><li>Operate the system and determine if the concern is still present.</li></ul> **Is the concern still present?** |

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test AN: DTC U3003:16**

**Diagnostic Overview**

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| U3003:16 | Battery Voltage: Circuit Voltage Below Threshold | Set by the ACM , APIM , FCIM , and GPSM when the supply voltage falls below 10 volts for at least 10 seconds during normal operation or for more than 250 milliseconds during the self-test. |

**PINPOINT TEST AN : DTC U3003:16**

| AN1 RECHECK FOR DTC U3003:16 |
|---|
| <ul><li>Ignition ON.</li><li>Using a diagnostic scan tool, clear the DTCs for the module in question.</li></ul> |

- Wait at least 15 seconds.
- Using a diagnostic scan tool, repeat the self-test for the module in question.

**Is DTC U3003:16 still present?**

| | |
|---|---|
| **Yes** | GO to AN2. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to a previous low battery voltage condition. |

**AN2 CHECK FOR CHARGING SYSTEM DTCS IN THE PCM**

- Using a diagnostic scan tool, perform the PCM self-test.

**Are any voltage-related DTCs set in the PCM ?**

| | |
|---|---|
| **Yes** | REFER to PCM DTC Chart in Section 303-14. |
| **No** | GO to AN3. |

**AN3 CHECK THE BATTERY CONDITION AND STATE OF CHARGE**

- Check the battery condition and verify that the battery is fully charged. REFER to Battery Condition Test in Section 414-01.

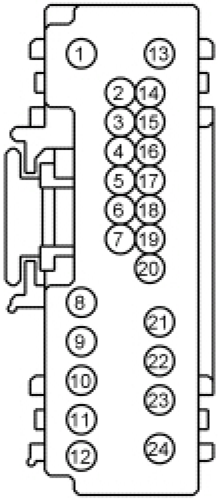
**Is the battery OK and fully charged?**

| | |
|---|---|
| **Yes** | GO to AN4. |
| **No** | REFER to the Symptom Chart in Section 414-00. |

**AN4 CHECK THE MODULE VOLTAGE SUPPLY**

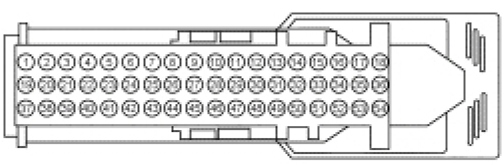
- Ignition OFF.
- Measure and record the voltage at the battery.
- Disconnect the audio system module setting DTC U3003:16.
- Ignition ON.
- Measure the voltage between the module setting DTC U3003:16 and ground.
- For the ACM , measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C290A Pin 1 | SBP29 (GY/RD) | — | Ground |

- 

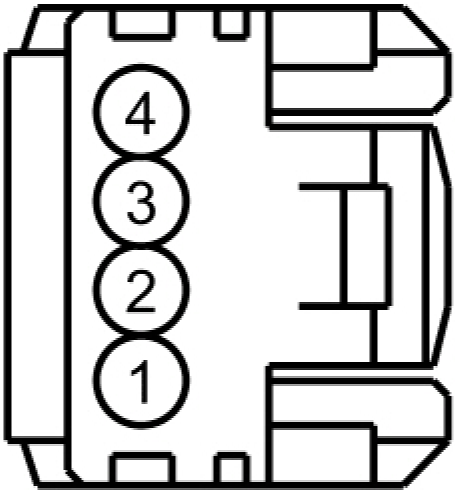

- For the APIM , measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C2383 Pin 1 | SBP02 (YE/RD) | — | Ground |

- 

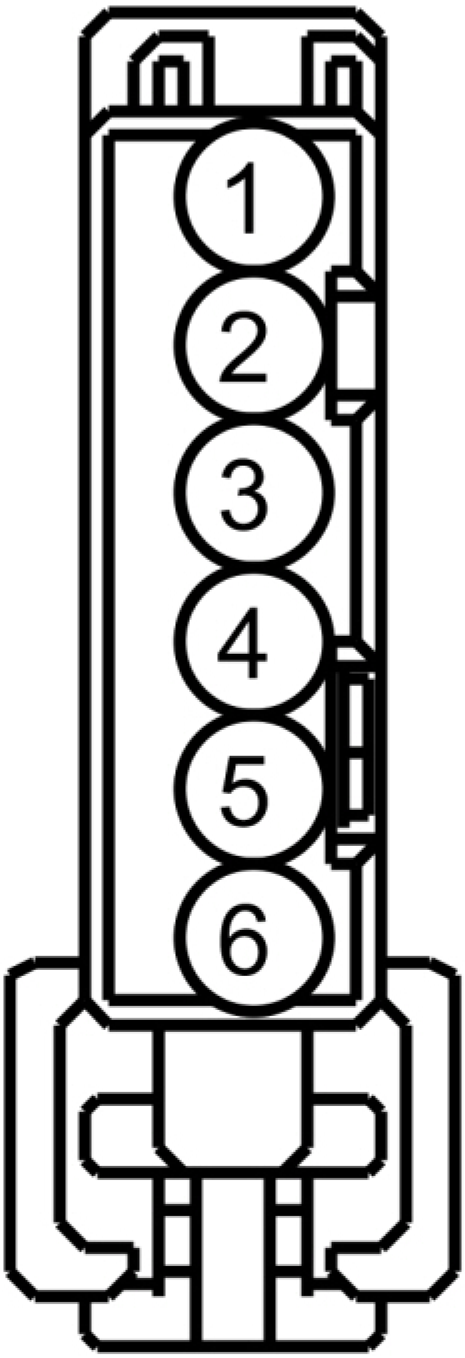
- For the FCIM , measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C2402 Pin 1 | SBP09 (RD) | — | Ground |

- 

- For the GPSM , measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C2398 Pin 1 | SBP09 (RD) | — | Ground |




- 

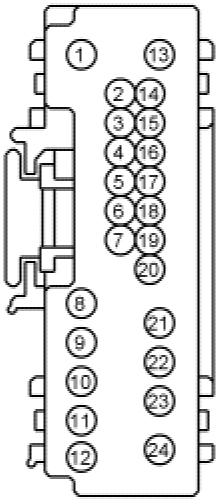
**Is the voltage within 0.2 volts of the recorded battery voltage?**

| | |
|---|---|
| **Yes** | GO to AN5. |
| **No** | REPAIR the circuit for high resistance. |

**AN5 CHECK THE SUSPECT MODULE GROUND CIRCUIT FOR CONTINUITY**

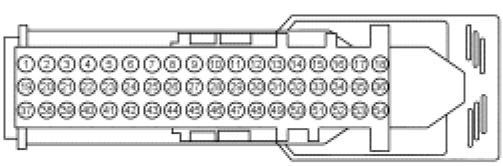
- Measure the voltage at the suspect module.
- For the ACM , measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C290A Pin 1 | SBP29 (GY/RD) | C290A Pin 13 | GD114 (BK/BU) |

- 

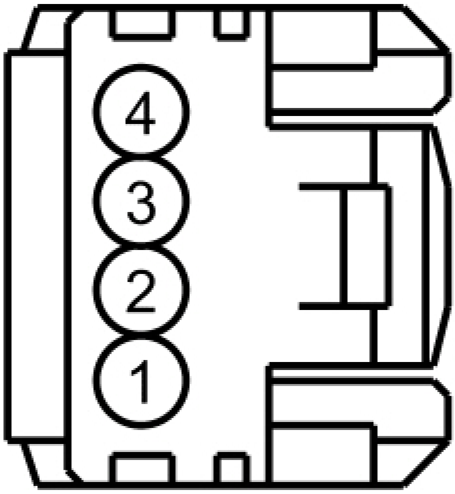

- For the APIM , measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C2383 Pin 1 | SBP02 (YE/RD) | C2383 Pin 37 | GD114 (BK/BU) |
| | | C2383 Pin 38 | GD114 (BK/BU) |

- 

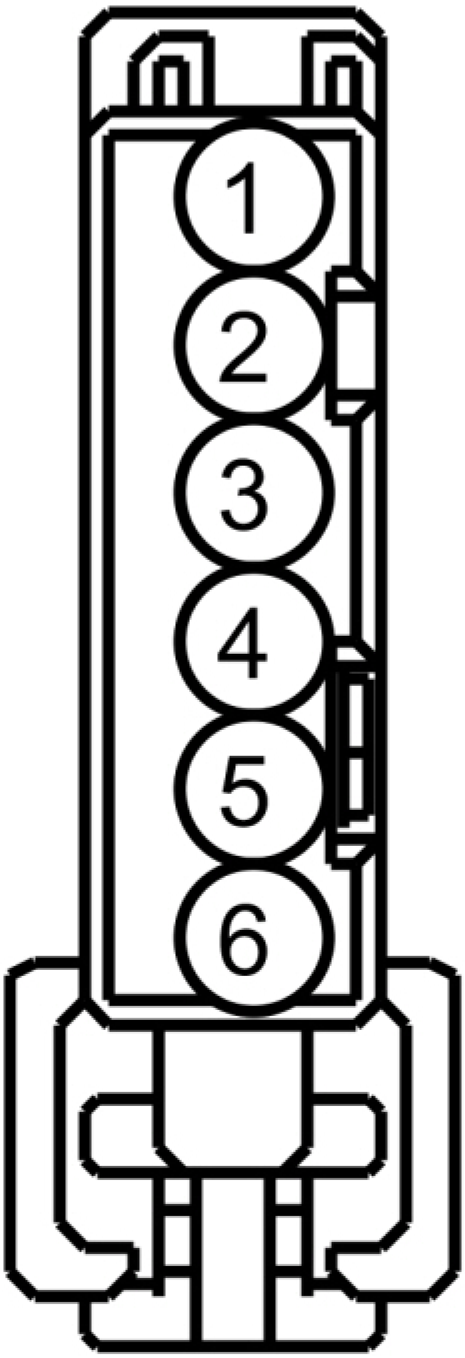
- For the FCIM , measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C2402 Pin 1 | SBP09 (RD) | C2402 Pin 4 | GD133 (BK) |

- 

- For the GPSM , measure the **voltage** between:

| Positive Lead | | Negative Lead | |
|---|---|---|---|
| Pin | Circuit | Pin | Circuit |
| C2398 Pin 1 | SBP09 (RD) | C2398 Pin 6 | GD114 (BK/BU) |




- 

**Is the suspect module voltage greater than 11 volts?**

| | |
|---|---|
| **Yes** | For the ACM , GO to AN6.<br>For the APIM , GO to AN7.<br>For the FCIM , GO to AN8.<br>For the GPSM , GO to AN9. |
| **No** | REPAIR the circuit for high resistance. |

**AN6 CHECK FOR CORRECT ACM OPERATION**

- Ignition OFF.

- Disconnect and inspect all the ACM connectors.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the ACM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**AN7 CHECK FOR CORRECT APIM OPERATION**

- Ignition OFF.
- Disconnect and inspect the APIM connector.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the APIM connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the APIM to re-initialize.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00. |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**AN8 CHECK FOR CORRECT FCIM OPERATION**

- Ignition OFF.
- Disconnect and inspect the FCIM connector.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)

- Pushed-out pins (install new pins as necessary)

- Reconnect the FCIM connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**AN9 CHECK FOR CORRECT GPSM OPERATION**

- Ignition OFF.
- Disconnect and inspect the GPSM connector.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the GPSM connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new GPSM . REFER to Global Positioning System Module (GPSM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

## Pinpoint Test AO: DTC U3003:17

### Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

### DTC Fault Trigger Conditions

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| U3003:17 | Battery Voltage: Circuit Voltage Above Threshold | Set by the ACM , APIM , FCIM , and GPSM when the supply voltage is more than 15.8 volts for more than 250 milliseconds during normal operation or the self-test. |

**PINPOINT TEST AO : DTC U3003:17**

**AO1 RECHECK FOR DTC U3003:17**

- Turn the engine off.
- Ignition ON.
- Using a diagnostic scan tool, clear the DTC for the module in question.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, perform the self-test for the module in question.

**Is DTC U3003:17 present?**

| | |
|---|---|
| **Yes** | GO to AO2. |
| **No** | The system is operating correctly at this time. The DTC may have been set previously during battery charging or while jump starting the vehicle. |

**AO2 CHECK FOR HIGH VOLTAGE DTCS SET IN OTHER MODULES**

- Ignition ON.
- Using a diagnostic scan tool, retrieve all CMDTCs .
- Wait at least 10 seconds.
- Using a diagnostic scan tool, perform the self-test for the module in question.

**Are high voltage DTCs such as B1676, P0563, U3003:17 set in more than one module?**

| | |
|---|---|
| **Yes** | REFER to Section 414-01 to diagnose an overcharging condition. |
| **No** | GO to AO3. |

**AO3 CHECK THE BATTERY VOLTAGE**

- Turn off all interior lights, exterior lights, and accessories.
- Start and run the engine at approximately 2,000 RPM for 3 minutes while monitoring the battery voltage.

**Does the battery voltage rise to 15.5 volts or higher?**

| | |
|---|---|
| **Yes** | REFER to Section 414-01 to diagnose an overcharging condition. |
| **No** | For the ACM , GO to AO4.<br>For the APIM , GO to AO5. |

| | |
|---|---|
| | For the FCIM , GO to AO6.<br>For the GPSM , GO to AO7. |

## AO4 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the ACM connectors.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the ACM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

## AO5 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the APIM connector.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the APIM connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the APIM to re-initialize.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00. |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

## AO6 CHECK FOR CORRECT FCIM OPERATION

- Ignition OFF.
- Disconnect and inspect the FCIM connector.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the FCIM connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |

**AO7 CHECK FOR CORRECT GPSM OPERATION**

- Ignition OFF.
- Disconnect and inspect the GPSM connector.
- Repair:
  - Corrosion (clean module pins or install new connectors or terminals)
  - Damaged or bent pins (install new terminals or pins)
  - Pushed-out pins (install new pins as necessary)
- Reconnect the GPSM connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new GPSM . REFER to Global Positioning System Module (GPSM). |
| **No** | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues. |